

Local Core Spacing and Filament Occupancy in *Herschel* Gould Belt Filaments[★]

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ABSTRACT

Dense cores along molecular-cloud filaments are often reported at separations shorter than the fastest-growing mode of an equilibrium isothermal cylinder, $\lambda_m \approx 5$ –6 filament widths. We re-analyse four *Herschel* Gould Belt Survey clouds with *Gaia* Data Release 3 distances and show that the size of this discrepancy, and to some extent its existence, depends on the statistic used.

The median of all pairwise separations is not a local-spacing estimator: it converges on a fixed fraction of the segment length it is applied to. Using adjacent along-crest separations, cores within populated stretches are separated by only ~ 1 –3 filament widths, whereas the filament length per core is 3.4–5.8: one cloud is consistent with the classical band, a second lies just below it, and two are sub-classical by up to a factor of two. The two diverge because cores occupy only 6–27 per cent of the crest, in tight groups separated by long empty stretches.

Core formation is therefore spatially intermittent rather than described by a single wavelength; the gap distribution is inconsistent with both a periodic comb and a Poisson process while showing no signature at the classical scale. The magnitude of the local deficit is set by the structural definition adopted, changing by a factor of two between a persistent-crest and a medial-axis network.

Simulations show that a contracting filament need not select the equilibrium wavelength, mode selection instead tracking the density at which the modes grow. They do not reproduce the observed point process and are offered as a counterexample, not an explanation.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established the ubiquity of filamentary structure in nearby star-forming clouds (André et al. 2010) and a characteristic width of ~ 0.1 pc across diverse environments (Arzoumanian et al. 2011), though the universality and physical interpretation of that scale remain contested (Panopoulou et al. 2017, 2022). For an isothermal cylinder the critical mass per unit length is $M_{\text{line,crit}} \approx 16 M_\odot \text{ pc}^{-1}$ (Ostriker 1964), and a marginally-stable cylinder fragments at $\lambda_m \approx 5$ –6 $\times$ FWHM (Inutsuka & Miyama 1992; Fischera & Martin 2012). Observed core separations are widely reported to fall below that value, and the deficit has been read as evidence for additional physics.

What this paper does. We re-measure core spacing along filament crests in four HGBS clouds with *Gaia* DR3 distances (Zhang 2023), and argue that the apparent conflict with classical theory is largely a question of which statistic is compared with which prediction.

[★] Results are established for the nearby, low-mass Gould Belt sample ($N = 4$ clouds) and are not offered as a universal fragmentation law.

Two observational points anchor the paper. First, the all-pairs (pairwise) median separation is not an estimator of local spacing at all. It converges to ≈ 0.293 times the extent of whatever segment it is applied to, so it measures the region selected (Section 2.6). We therefore use adjacent along-crest separations for local spacing and retain the all-pairs statistic only for historical comparison. We make no claim that published HGBS spacings generally suffer from this: the values we could check are local or along-fibre statistics that agree with ours (Table 12).

Second, and more importantly, the local and the global statistics give different answers, and that difference is the result. Measured only within core-bearing stretches, adjacent separations are several times shorter than λ_m under every skeleton and width convention we tried. Measured over the whole crest, the filament length per core is close to classical in two of the four clouds. The reconciliation is that cores occupy only 6–27 per cent of the crest: they form in tight groups separated by long bare stretches. The scientific question therefore shifts from “why is the fragmentation wavelength too short?” to “why is core formation spatially intermittent along the filament?”.

Candidate explanations. Three mechanisms could shorten

a local spacing. (i) *Hierarchical fragmentation*, in which filaments are bundles of velocity-coherent fibres (Hacar et al. 2013, 2018; Yang et al. 2024), so that measuring across the bundle compresses the apparent filament-level spacing. (ii) *Magnetic tension* (Nakamura et al. 1993): only a helical or toroidal field shortens the axial mode, since a static longitudinal field leaves it unchanged and a perpendicular field lengthens it. (iii) *Dynamical fragmentation*, in which a still-collapsing filament fragments at a scale set by its current contracting state rather than by the marginally-stable equilibrium mode; this is already qualitatively expected in the dynamical-ISM paradigm (André et al. 2014; Clarke et al. 2016, 2017, 2020).

The evidence here is not evenly distributed among them. We test (iii) directly, treat (ii) only with a geometric toy model, and do not test the force-balanced helical configuration at all, so (ii) is mentioned for completeness as an unconstrained alternative rather than as a coequal candidate. The data cannot discriminate between the three, and nothing below should be read as identifying the cause.

The question this paper asks. Stated in the form we think the data actually address it: *is core formation along molecular-cloud filaments adequately described by a single fragmentation wavelength, or is it intrinsically spatially intermittent?* The classical-cylinder comparison then becomes one diagnostic of that more general question rather than the organising principle, and S_{local} , S_{global} , the occupancy fraction, the Fano factor, the gap distribution and the pair-correlation function become different projections of the same underlying point process.

New contributions. These are three: a homogeneous, *Gaia*-DR3, multi-convention measurement of adjacent-core spacing and filament length per core across the four primary clouds, together with the occupancy statistics that connect them; the identification of the region-extent bias in all-pairs separation statistics in this context (the underlying triangular-distribution order statistic is elementary and we claim no new mathematics); and a controlled numerical experiment in which the identical pipeline recovers the classical wavelength for a relaxed equilibrium cylinder but finds no resolved maximum for a contracting one, extended here to a five-point ladder in contraction rate.

Scope. With four independent clouds these are sample statements, not a universal fragmentation law, and all $\pm$ values quoted in this paper are descriptive $N = 4$ ranges rather than confidence intervals. Symbols are collected in Table 1.

Notation. Four normalised quantities carry the argument. Capital symbols denote ratios to the classical benchmark; $s_{\text{ACS,med}}$ denotes a raw separation in parsecs.

- $S_{\text{global}} = (L/N)/W$, the unconditional filament length per core. For a *continuously unstable* filament producing approximately one core per unstable wavelength, this is the corresponding observational statistic, and it is the natural comparison with $\lambda_{\text{m}}/W \approx 5\text{--}6$. The correspondence is exact only if the whole measured skeleton was unstable, which it is not (Section 2.9); three lengths must therefore be distinguished, L_{skel}/N , $L_{\text{core-bearing}}/N$ and L_{unstable}/N , of which only the last maps directly onto the ideal cylinder.

- $S_{\text{local}} = s_{\text{ACS,med}}/W$, the median adjacent-core separation measured *only within core-bearing stretches*: a descriptor

Table 1. Principal symbols.

Symbol	Meaning
$s_{\text{ACS,med}}$	observed adjacent-core spacing (ACS) statistic (median adjacent gap; a line-density scale, <i>not</i> a resonant/fragmentation wavelength; cf. Section 2.12)
W_{fil}	observed filament FWHM (beam-deconvolved Plummer; the reference width)
W_{core}	simulation initial Gaussian half-width σ ($0.3 \lambda_{\text{J}}$)
W_{form}	FWHM of the evolved, projected simulation profile
$T_1 \equiv \eta_W$	$W_{\text{form}}/W_{\text{fil}}$, the simulation-to-observation width normalisation (forward-model band 0.59–0.78; adopted 0.61)
λ_{J}	Jeans length at the <i>initial background</i> density ρ_0 ($= 1$ in code units) and never at the evolving central density, for which we write $\lambda_{\text{J}}(\rho_c)$
λ_{peak}	wavelength at which $\Gamma(\lambda)$ is maximal. Resolved <i>only</i> for the relaxed equilibrium control (at $22H$); <i>not</i> resolved for any contracting run, for which we quote λ_{turn} instead
λ_{turn}	upper bound on where a turnover in $\Gamma(\lambda)$ could lie for a contracting filament ($\lesssim 1.1 \lambda_{\text{J}}$); the only wavelength we quote for those runs
$\lambda_{\text{m}} \equiv \lambda_{\text{IM92}}$	classical Inutsuka–Miyama equilibrium-cylinder wavelength $\approx 5\text{--}6 W_{\text{FWHM}}$ ($\approx 4 \times$ flat diameter)
$\Gamma(\lambda)$	linear growth rate of axial mode of wavelength λ
f	thermal line-mass fraction $M_{\text{line}}/M_{\text{line,crit}}$
β	plasma beta $8\pi P/B^2$
$\mathcal{M}$	amplitude label of the seed perturbation ($\delta v = \mathcal{M}c_s \times 10^{-4}$; a linear seed, <i>not</i> a turbulent Mach number; driven transonic turbulence is tested separately in the deposited material)
$\mu_{\Phi}/\mu_{\Phi,\text{crit}}$	normalised mass-to-flux ratio
θ	angle between field and filament axis
N_{J}	cells per local Jeans length (Truelove criterion)
C	density contrast $\rho_{\text{max}}/\langle \rho \rangle$

of the interior of core-bearing regions, not a fragmentation wavelength.

- $S_{\text{elig}} = (L_{\text{elig}}/N_{\text{elig}})/W$, S_{global} restricted to filament above the critical line mass; an exploratory diagnostic (Section 2.9).

- $S_{\text{sim}}^{\text{upper}} = \lambda_{\text{turn}}/W$, an *upper bound* from the simulations, never a measured wavelength.

The difference between S_{global} and S_{local} is itself the measurement of how unevenly cores occupy the crest. Every other ratio quoted here is a sensitivity test or alternative convention on one of these four, decoded in Table 4.

2 OBSERVATIONAL ANALYSIS

2.1 The sample

Our sample comprises 8 high-confidence HGBS regions (four *primary*, Table 2; four *limited*, Appendix A). The original catalogues (Könyves et al. 2015; Könyves et al. 2020; Ladjelate et al. 2020; Pezzuto et al. 2021) adopted distances of 130–400 pc from the best estimates available in 2010–2015

Table 2. Primary HGBS sample with Gaia DR3 distances. Per-region spacings are pairwise-median values rounded to three significant figures; the core-count-weighted mean of the four primary regions is 0.290 pc. The four *limited* regions (Ophiuchus, Serpens, TMC1, CRA) and the full 8-region weighted mean (0.279 pc) are listed in Appendix A, the deposited material.

Region	D (pc)	N_{core}	Pairwise med. (pc)	S.E. (pc)	Status
Orion B	386	1,844	0.313	0.047	Primary
Aquila	436	749	0.346	0.047	Primary
Perseus	300	816	0.248	0.040	Primary
Taurus	135	536	0.198	0.040	Primary
Weighted Mean[†]		3,945	0.290		

[†]Core-count-weighted mean of the four primary regions. A formal core-count-weighted standard error (treating cores as independent) is *not* the relevant uncertainty, since the four clouds are the independent units (Section 2.13); the cloud-to-cloud scatter is carried instead.

(Table 12 lists the pre-Gaia distance and its source per region; note that Könyves et al. 2020 already adopted ~ 400 pc for Orion B); Gaia DR3 parallaxes (Zhang 2023) have since enabled substantial revisions, largest for the more distant regions, whose pre-Gaia distances were the most uncertain.

We adopt the Gaia DR3 YSO-based distances from Zhang (2023) (Table 4), with typical uncertainties of 3%–5%. The revisions are strongly region-dependent (largest for Serpens 260 $\rightarrow$ 458 pc, +76%, and Aquila 260 $\rightarrow$ 436 pc, +68%; Orion B, Taurus and Ophiuchus essentially unchanged); since spacings scale linearly with distance, the core-count-weighted mean rises only modestly ($\sim 10\%$ for the primary regions), dominated by Aquila rather than being a uniform +32% shift. Because Gaia DR3 thus compresses or expands each cloud non-uniformly, the revision is not a single global rescaling, which is a further reason the clouds must be treated as $N = 4$ independent units rather than by pooling cores that carry very different distance corrections. The observed spacing is a line-density statistic, not a resonance, and coincides with the theoretical wavelength only to order of magnitude.

The analysis includes all HGBS-identified cores (starless, prestellar and protostellar), extracted and classified within the consistent HGBS framework (Könyves et al. 2015; Könyves et al. 2020). Including unbound starless and protostellar sources may introduce some bias, but fragmentation produces a continuum of boundness states, and restricting the sample to prestellar cores would shrink several regions (notably TMC1, CRA and Serpens) below statistical usefulness. Core positions and properties are taken directly from the published catalogues.

Protostellar migration displacements of 0.01–0.05 pc (Kirk et al. 2016; Mattern et al. 2018) are small against the ~ 0.2 – 0.3 pc spacing; we estimate the resulting ACS bias at ~ 5 – 10% and carry it as a systematic (Section 2.13).

2.2 Measurement method

The analysis pipeline, from raw HGBS maps to the final characteristic $s_{\text{ACS,med}}/W$, proceeds from the HGBS column-density maps and Gaia DR3 distances through skeleton extraction, morphological bridging, core association within 0.06 pc of a spine, graph ordering along each component and

adjacent-core separation. Filament skeletons were identified using DisPerSE (Sousbie 2011) following standard HGBS methodology (Arzoumanian et al. 2011), with a 3σ persistence threshold and manual editing restricted to removing obvious artefacts (e.g. spurs connecting unrelated structures). Its effect on the spacing is small ($\lesssim 5\%$), bounded by the junction-retention and association-threshold sensitivity tests as a *proxy* (Table 9). Because the skeleton algorithm is the dominant systematic in this paper, robustness to the persistence threshold of the adopted ontology is part of the measurement and not future work, and we have therefore tested it in two ways (Appendix D). The manual editing itself is bounded only by these proxies and is carried as such in the error budget (Section 2.13). Because skeletons and core positions derive from distance-independent angular positions, the Gaia DR3 revisions require no re-extraction. The measured spacing is insensitive to these choices: excluding junctions changes it by $< 5\%$ and varying the association threshold between $1\times$ and $1.5\times$ the width by $< 3\%$ (a coarse two-point check, distinct from the finer 0.04–0.08 pc half-width sweep of Section 2.7 that gives a 4–7% per-region spread; the primary ACS pipeline associates cores within a half-width, ~ 0.06 pc, of a spine).

Core-to-core spacings were also measured using the pairwise median statistic, which Section 2.6 shows converges to $\approx 0.293L$ rather than the fragmentation wavelength for $N \gtrsim 5$; we retain it only as a comparability statistic with prior HGBS work.

Because physical spacing is angular separation times distance, the $\lambda \propto d$ scaling is tautological and we do not present it as corroboration; the relevant check is that the *residuals* about that scaling show no trend with the absolute revision fraction $|\Delta D/D_{\text{original}}|$, the two most-revised clouds (Aquila, Serpens) do not stand out as having the largest or smallest residuals, so the revision does not preferentially bias the most-revised clouds. With only four clouds a formal correlation test has negligible power and we do not quote one. Two caveats remain. The YSO-based distances (Zhang 2023) are assumed to apply to the dense gas measured here, and the Aquila/Serpens Rift has substantial line-of-sight depth ($\sim$ tens of pc in the 3-D dust reconstructions of Zucker et al. 2020 and Green et al. 2024), so a single assigned distance may blend populations. We quantified the blending with a geometric Monte-Carlo model in which two crests at different depths are projected onto a common spine, taking an intrinsic spacing of 0.20 pc at $d = 436$ pc. The model draws each crest’s line-of-sight offset from a Gaussian of dispersion σ_z , with the two crests separated by $2\sigma_z$ in depth, and we scan $\sigma_z = 5, 10, 15$ and 30 pc (total separations of 10–60 pc, spanning the $\sim$ tens of parsecs quoted by Zucker et al. 2020 and Green et al. 2024 for the Rift). Superposition interleaves the two chains and *shortens* the measured median, to 0.41–0.49 of the true value across that range, whereas the distance spread alone (which rescales each spacing by $d/(d+z)$) contributes only ± 1 to ± 7 per cent. Blending therefore biases the ACS towards *more* sub-classical values rather than less. The ~ 50 per cent figure is a worst case in which two crests overlap completely; the realistic effect is smaller because most associated cores lie on coherent crests. We carry it as a one-sided systematic in Table 9. This is a geometric model with a free depth parameter, not a measurement. The stronger treatment, projecting the three-dimensional dust reconstructions

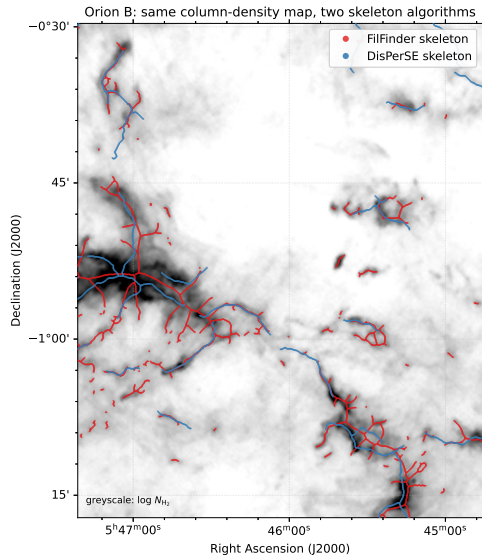


Figure 1. The dominant observational systematic, shown directly: the DisPerSE (blue) and FilFinder (red) skeletons extracted from the *same* *Herschel* column-density map of Orion B (greyscale, $\log N_{\text{H}_2}$; a representative sub-region). The FilFinder skeleton is produced by FILFINDER2D with the cloud distance set per region and an *externally supplied* mask (`use_existing_mask=True`) followed by `medskel()`; the mask is a simple column-density cut at the 92nd percentile of the finite pixels, which corresponds to $N_{\text{H}_2} = 2.5, 3.8, 3.1$ and $2.9 \times 10^{21} \text{ cm}^{-2}$ in Orion B, Aquila, Perseus and Taurus respectively. Because the mask is supplied externally, FILFINDER’s own adaptive and branch thresholds are not invoked. A more permissive 70th-percentile mask is also computed and is what sets the upper end of the FilFinder range. The DisPerSE skeletons are the published HGBS products, extracted at a 3σ persistence cut. The two algorithms trace substantially different filament networks, FilFinder’s medial axis of a column-density mask is denser and more branched and follows more low-contrast material, whereas DisPerSE follows the persistent crests, which is why the inferred $s_{\text{ACS,med}}/W$ differs by a factor of ~ 2 between them (Section 2.13). This is intrinsic to how each algorithm defines a “filament” and is the largest single uncertainty on the *magnitude* of the spacing, though the sub-classical *sign* holds for both.

of Zucker et al. (2020) and Green et al. (2024) along the actual sightline through the Rift and counting how many physically distinct crests fall within one association length, is beyond the scope of this paper and is listed as a specific follow-up in Section 6.4. As a floor, retaining the pre-Gaia distances lowers the primary-region characteristic to $s_{\text{ACS,med}}/W \approx 1.4$ (still sub-classical), so the sub-classical *sign* does not depend on the distance revision, though its magnitude does.

The measurement convention, above all the skeleton algorithm, dominates the numerical value: the DisPerSE and FilFinder skeletons of the identical Orion B map (Fig. 1) trace substantially different networks, moving the inferred $s_{\text{ACS,med}}/W$ by a factor ~ 2 while the sub-classical *sign* holds for both (Section 2.13).

2.3 Results

The four primary regions are the independent statistical units throughout. Individual cores and spacings within a cloud are

Table 3. Principal results, one row per cloud. $S_{\text{local}} = s_{\text{ACS,med}}/W$ is the conditional adjacent-core spacing, measured only within core-bearing stretches; $S_{\text{global}} = (L/N)/W$ is the unconditional filament length per core; $S_{\text{elig}} = (L_{\text{elig}}/N_{\text{elig}})/W$ restricts the latter to filament above the critical line mass, selected on a core-masked column-density reconstruction (Section 2.9), with f_{elig} the eligible length fraction and N_{elig} the cores on it; f_{occ} is the 0.06 pc occupancy fraction. All are normalised by the fixed $W = 0.10$ pc width so the columns are directly comparable, and all use the DisPerSE skeleton. The classical benchmark is 5–6. The simulated upper bound is common to all rows, $S_{\text{sim}}^{\text{upper}} = \lambda_{\text{turn}}/W \lesssim 1.0$ –1.3 (Section 4); it is an upper bound, not a measured wavelength. The Perseus entry is bracketed because only 57 cores remain on eligible filament after masking and the value depends on the masking radius (Table E1); no value is quoted for Taurus, where 13 cores remain and the statistic does not stabilise.

Cloud	S_{local}	S_{global}	S_{elig}	$f_{\text{elig}} [N_{\text{elig}}]$	f_{occ}
Orion B	1.8	3.4	2.0	0.16 [268]	0.27
Aquila	1.7	3.4	3.2	0.90 [303]	0.24
Perseus	3.0	5.8	(4.1)	0.08 [57]	0.11
Taurus	1.8	4.5	—	— [13]	0.06

not independent, and the nesting is propagated in the hierarchical bootstrap of Section 2.13.

Two statistics, two different answers. The adjacent-core spacing $s_{\text{ACS,med}}$ is *conditional*: measured only where cores are adjacent, it characterises the interior of core-bearing stretches and says nothing about filament that has formed no cores. The core line density N/L , quoted through its reciprocal L/N , is *unconditional*. For a filament that is unstable along its whole measured length and produces one core per unstable wavelength, L/N is the corresponding observational statistic. That identification is exact only under those conditions: the measured skeleton length is not in general the length of unstable cylinder to which the theoretical prediction applies, and a filament half of whose length is subcritical will return $L/N \simeq 2\lambda$ even if the unstable half fragments perfectly at λ . Segmentation therefore changes L/N without changing the fragmentation physics, which is why S_{global} moves from 3.4–5.8 to 9.1–16.7 between the two skeleton definitions (Section 2.10) and why the eligible-length diagnostic of Section 2.9 is introduced immediately below.

Taking the predicted quantity first, $S_{\text{global}} = 3.4, 3.4, 4.5$ and 5.8 for Orion B, Aquila, Taurus and Perseus. *On the comparison classical theory actually makes, two of the four clouds are therefore not discrepant at all*, and the other two fall below the classical band by less than a factor of two. Conditionally, S_{local} is sub-classical by a factor ≈ 2.4 –4.3 (Table 4). The reconciliation is the occupancy result of Section 2.11: cores occupy only 6–27 per cent of the crest.

Every comparison of S_{local} with 5–6 W below, including Fig. 2, is therefore a comparison between a conditional descriptor and a fragmentation wavelength. It is not a measurement of a shortened fragmentation wavelength and we do not use it as one. The sign of the local deficit is robust under every convention we tried; its magnitude is not, being dominated by the choice of skeleton. Figure 7 shows the full methodological envelope on a single axis and is the quickest orientation to the sections that follow (Section 2.17).

Table 4. Convention decoder for the *conditional* local descriptor: read every observed $s_{\text{ACS,med}}/W$ quoted in the text against it. This table tabulates the adjacent-core spacing, which is measured only within core-bearing stretches and is *not* the quantity a one-fragment-per-wavelength theory predicts (that is L/N ; Section 2.8). Values are given under each width/skeleton convention and correction state for the four primary HGBS regions (revised distances); every combination is conditionally sub-classical (classical $\lambda_m \approx 5\text{--}6 \times \text{FWHM}$). We adopt the directly-measured region-Plummer value as our reference, $(s_{\text{ACS,med}}/W)_0 \approx 1.4$; the fixed-width value (≈ 1.9 , retained for comparability with prior HGBS work) and all others are read against it. The “periodic” column applies the periodic injection–recovery correction to the raw value; the data-matched *clustered* column (bracketed) applies $\div 1.64$. The DisPerSE conventions are sub-classical by a factor $\approx 2.4\text{--}4.3$; across the full envelope the factor spans $\approx 2\text{--}13$, the upper end set by the densest FilFinder skeleton.

Quantity	raw	periodic
<i>Conditional local descriptor</i> $s_{\text{ACS,med}}/W$ (core-bearing stretches only)		
Region Plummer, DisPerSE (adopted, $(s_{\text{ACS,med}}/W)_0$)	1.4	1.2
Fixed 0.10 pc, DisPerSE	2.1	1.9
FilFinder skeleton	0.4–0.8	0.4–0.8
Per-segment local width, $\Delta s/W_{\text{local}}$		0.6–1.1
<i>Deprojected</i> (adopted row $\div (\sin i)_{\text{med}} = 0.87$)	1.6	1.4
<i>Deprojected</i> (fixed 0.10 pc)	2.4	2.2
<i>Unconditional</i> (what fragmentation theory predicts)		
Global length per core $(L/N)/W$, fixed 0.10 pc		3.4–5.8
Mean adjacent, fixed 0.10 pc (branch-jump inflated)		2.1–4.7
<i>Legacy comparator and reference scales</i>		
Pairwise median ($\approx 0.293L$; biased, not a spacing)	2.8	
Matched simulation (upper limit)	$\lambda_{\text{turn}}/W \lesssim 1.0\text{--}1.3$	
Classical λ_m (F&M 2012)	5–6 (= $4 \times \text{diam.}$)	

Notes The reference point is the region-Plummer ACS ($s_{\text{ACS,med}}/W)_0 \approx 1.4$ (bold), one illustrative point within the convention range 0.8–1.9; it is an adopted reference value, chosen for comparability with prior HGBS work, not a uniquely preferred physical scale (and not a formal confidence interval; Section 2.13). The data-matched *clustered* branch ($\div 1.64$ on the raw value, giving ≈ 0.8 region-Plummer, ≈ 1.2 fixed-width and $\approx 0.3\text{--}0.5$ FilFinder) is *not* shown as a column because it is not independent: it applies a correction derived by *assuming* the observed clustering (Section 2.7), so it is circular and serves only as a lower bound; we carry the raw value throughout. The fixed-width convention gives a *higher* value (≈ 1.9) because it uses 0.10 pc rather than the larger measured widths; FilFinder gives lower values still. The full envelope factor $\approx 2\text{--}13$ (Section 2.7) is $\lambda_m/(s_{\text{ACS,med}}/W)$: its lower endpoint ≈ 2 uses $(\lambda_m, s_{\text{ACS,med}}/W) = (5.5W, 2.6)$ (fixed-width DisPerSE, Perseus) and its upper endpoint ≈ 13 uses $(5.5W, 0.42)$ (densest FilFinder skeleton). We quote $s_{\text{ACS,med}}/W$ to one decimal place; finer digits are not significant. The two *deprojected* rows apply the median isotropic projection factor $(\sin i)_{\text{med}} = 0.87$ (Section 2.13); this correction acts *against* our conclusion, raising the ratio by 15–27 per cent, and we show it in the table rather than only in the text so that the effect is visible in the same place as the adopted numbers. We nevertheless carry the uncorrected value as the headline, because deprojection assumes an isotropic orientation distribution that the filaments in a single cloud need not obey.

2.4 The dominant observational systematic: filament width

Two width conventions recur throughout: the *fixed-width* convention (the canonical HGBS $W_{\text{fil}} = 0.10$ pc for every region) and the *region-Plummer* convention (a beam-deconvolved Plummer FWHM fitted per region); since $s_{\text{ACS,med}}/W \propto 1/W$, the broader region-specific widths give a *lower* $s_{\text{ACS,med}}/W$. The fixed $W_{\text{fil}} = 0.10$ pc convention is the largest single lever on $s_{\text{ACS,med}}/W$, comparable to the whole discrepancy, so we measured region-specific widths directly. Along each spine we sampled the column-density profile perpendicular to the local tangent and fitted a beam-deconvolved Plummer function (the HGBS width definition; Arzoumanian et al. 2019) to hundreds of profiles per region (18.2'' beam; Table 5). All four regions give $p \simeq 2.0$ and $W_{\text{fil}} = 0.11\text{--}0.16$ pc, agreeing with the published HGBS values and slightly broader than 0.10 pc, so the region-specific $s_{\text{ACS,med}}/W$ (1.1–1.8, median ≈ 1.4) is *lower* than the raw fixed-width value (≈ 2.1 ; Table 4); every region is sub-classical under either convention. A Gaussian-crest estimate returns a much narrower Taurus width (0.031 pc). We reject it on physical grounds rather than because of the ratio it implies: a Gaussian fits only the inner ridge and ignores the power-law wings that carry most of the column density, whereas the Plummer form models the wings and is the profile used to define the HGBS filament width. The Plummer fit (0.11 pc, $s_{\text{ACS,med}}/W \approx 1.1$) makes Taurus the most sub-classical region. The result is thus not conditional on the width convention. We prefer the Plummer widths on physical grounds: they fit the observed $p \simeq 2$ profile of HGBS filaments and are far less sensitive to the fitting radius than either a single fixed 0.10 pc value or a Gaussian inner-ridge fit.

The fit is consistent across the sample: every region returns a Plummer index $p \simeq 2.0$, the value found for HGBS filaments generally (Arzoumanian et al. 2019), and a width within $\sim 50\%$ of 0.10 pc (inter-quartile ranges $\sim 0.07\text{--}0.20$ pc), so the sub-classical $s_{\text{ACS,med}}/W$ is a property of all four primary regions, not of one anomalous cloud. The single earlier outlier, a spuriously narrow 0.031 pc Gaussian crest for Taurus, was a profile-selection effect of fitting the inner ridge rather than the broad Plummer wings (not a pixel-scale artefact: the 3'' pixels subtend 0.002 pc at 135 pc); it does not survive the Plummer fit, which gives 0.11 pc and $s_{\text{ACS,med}}/W \approx 1.1$.

2.5 Sample classification and distance robustness

We classify the eight regions (Table 2; Appendix A) as **primary** (Orion B, Aquila, Perseus, Taurus) or **limited** by three criteria fixed *a priori* (before the measurement): $N > 500$ cores; a well-established distance; and a skeleton simple enough for unambiguous core-to-filament association. This selection is cross-checked by the leave-one-out test below. The limited sample fails at least one criterion (Ophiuchus on skeleton complexity, the others on small samples) and are excluded from the primary ACS measurement. The largest distance revisions (Serpens +76%, Aquila +68%) are independently corroborated by VLBI maser parallaxes (validating Gaia DR3 to $< 5\%$; Kounkel et al. 2017; Xu et al. 2019; Ortiz-León et al. 2018, 2017), extinction estimates (Yan et al. 2022; Green et al. 2024) and co-spatiality with Aquila

Table 5. Region-specific, beam-deconvolved filament widths from a direct Plummer fit to perpendicular column-density profiles along the skeletons (the HGBS width definition; Arzoumanian et al. 2019), and the resulting ACS $s_{\text{ACS,med}}/W$. W_{fil} is the beam-deconvolved Plummer FWHM (18.2'' beam), p the Plummer index (a free fit parameter, not fixed; it converges to $p \simeq 2.00$ in all four regions (Taurus 2.01, a difference in the second decimal place only), consistent with the canonical Plummer-2 profile), N the number of fitted profiles. Quoted W_{fil} uncertainties are the standard error on the median of the per-profile fits within each region and do not include the beam-deconvolution or background-subtraction systematics, which are carried separately in the error budget (Section 2.13); the fitted p has a per-region dispersion of ± 0.1 across profiles. All four regions give $p \simeq 2$ and sub-classical $s_{\text{ACS,med}}/W$ on the (primary-sample) median spacing quoted here; the mean adjacent spacing is a factor ≈ 1.3 –2.2 larger (Section 2.12), so the mean-based ratio is ≈ 2.1 –4.7, still sub-classical in every region.

Region	W_{fil} (pc)	p	N	ACS (pc)	$s_{\text{ACS,med}}/W$
Taurus	0.110	2.01	295	0.124	1.1
Orion B	0.148	2.00	356	0.207	1.4
Perseus	0.147	2.00	235	0.263	1.8
Aquila	0.161	2.00	446	0.210	1.3
median	0.148	2.00	,	,	1.4

The measured widths (0.11–0.16 pc) agree with the published HGBS values and exceed the canonical 0.10 pc, so region-specific Plummer widths give a lower $s_{\text{ACS,med}}/W$ (≈ 1.4 , raw) than the raw fixed-width value (≈ 2.1 ; Table 4): all four regions are sub-classical under either convention. ACS spacings are the full-coverage medians (Section 2.7), measured on the same DisPerSE skeleton as the widths (its deposited crest for the width fit, its bridged form for the spacing); the injection–recovery bias correction lowers each $s_{\text{ACS,med}}/W$ by a further ~ 10 –20%. (Pipeline outputs retained to 2 s.f.; the convention systematic dominates.)

(Zucker et al. 2020); Serpens is *limited* and excluded in any case. The sub-classical conclusion is independent of these regions: leave-one-out shifts the weighted mean by $< 6\%$, and dropping Aquila leaves the region-Plummer median at 1.40 and the fixed-width characteristic at 2.1, both sub-classical. The primary/limited split is likewise not tuned to the distance revision: its criteria act on the distance-independent core count and skeleton morphology, the closest any region lies to the $N > 500$ boundary is Taurus at 536 cores ($\sim 7\%$ above), and because the Gaia revision rescales spacings but not core counts, no plausible distance change moves a region across the classification.

2.6 Choice of spacing statistic

We adopt the along-skeleton adjacent-core spacing (ACS), denoted $s_{\text{ACS,med}}$ for its per-region median, as the primary local measure, and report the all-pairs median only for comparability with prior work. It is not the Euclidean nearest-neighbour distance, which we use only for the Clark–Evans statistic of Section 2.12.

An all-pairs median, used as a local-spacing estimator, is intrinsically dominated by the extent of the selected segment. For N points distributed over a segment of length L it converges to $d^* = L(1 - 1/\sqrt{2}) \approx 0.293L$, the median of the triangular pair-distance law, for a random and a perfectly periodic distribution alike (Appendix B). It tracks that value for any filament with $N \gtrsim 5$ cores and never approaches the

local spacing, which is 30–100 times smaller. The order statistic is elementary; what matters here is its consequence for a statistic in common use.

Synthetic point processes confirm this directly. For cores placed by periodic, clustered (gamma-renewal), hierarchical (three projected fibres) and fractal processes with a fixed injected local scale, the ACS median is flat with segment extent and recovers each process’s local scale, whereas the pairwise median rises as $0.29L$ for *every* process, independent of the injected scale.

We reconstruct the along-skeleton core ordering for the four primary regions from the deposited DisPerSE skeleton maps, bridging fragmented segments by morphological closing (Section 2.2). Figure 2 shows the per-region spacings.

2.7 Injection–recovery validation

To test whether the estimator recovers a known spacing under realistic conditions, synthetic cores with periodic spacings of 0.15–0.35 pc were placed on HGBS-property-matched skeletons and passed through the identical pipeline. The recovery is biased low by 11–29 per cent, because bridging and association occasionally split or merge a pair, and the correction is applied as a per-region multiplicative factor.

The size of that correction depends on the assumed point process, which matters because the real cores are not periodic. Repeating the injection with a clustered (gamma-renewal) process matched to the observed coefficient of variation gives a larger correction ($\div 1.64$). That branch is not independent, since it derives the correction by assuming the clustering it is meant to test, so we carry the periodic-injection value throughout and quote the clustered branch only as a lower bound (Table 4). Sweeping the association half-width over 0.04–0.08 pc changes the per-region median by 4–7 per cent.

Full-coverage measurement. The measurement uses the dense deposited DisPerSE skeletons for all four primary regions. As a segmentation sensitivity test rather than an independent confirmation, since it uses the same maps and catalogue, we repeated the whole analysis on FilFinder skeletons (Section 2.10).

2.8 Core line density and the local-width test

The median adjacent spacing measures the separation *where cores are adjacent*. For a filament unstable along its whole measured length and producing one core per unstable wavelength, the corresponding statistic is the core line density N/L , quoted through its reciprocal L/N . That identification holds only under those conditions: the measured skeleton length is not in general the length of unstable cylinder to which the prediction applies, and a filament half of whose length is subcritical returns $L/N \simeq 2\lambda$ even if the unstable half fragments perfectly at λ . Segmentation therefore changes L/N without changing the physics, which is why S_{global} moves from 3.4–5.8 to 9.1–16.7 between the two skeleton definitions (Section 2.10), and why the eligible-length diagnostic of Section 2.9 follows immediately below.

We report three statistics, all normalised by the same fixed 0.10 pc width. (i) The median adjacent spacing is $S_{\text{local}} \approx 1.7$ –3.0, sub-classical in every cloud. (ii) The mean adjacent

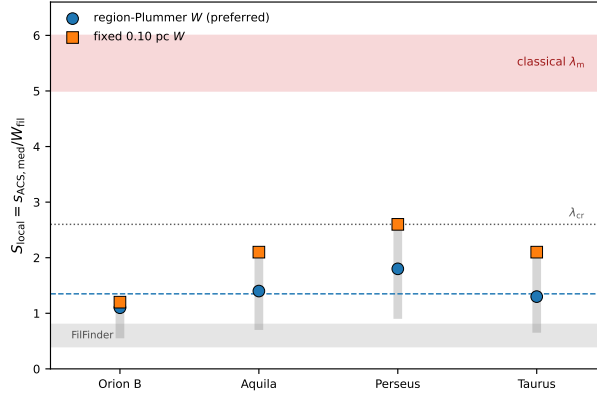


Figure 2. The conditional local descriptor $S_{\text{local}} = S_{\text{ACS,med}}/W_{\text{fil}}$ per primary region. This is *not* the quantity classical theory predicts (that is S_{global} ; Fig. 4b) and the comparison with the classical band below should be read diagnostically. It is the *conditional* descriptor measured within core-bearing stretches; the unconditional length per core L/N , whose reciprocal is the line density predicted by fragmentation theory, is shown in Fig. 4, in the two DisPerSE width conventions: directly-measured region-Plummer widths (blue circles, preferred; per-region raw 1.1, 1.4, 1.8, 1.3, dashed line at the median, which we quote as 1.4, the rounded value of the exact median of 1.35) and the fixed 0.10 pc width (orange squares; 1.2, 2.1, 2.6, 2.1), both shown *raw* (before the injection–recovery correction; Section 2.7). The red band is the classical fastest-growing wavelength $\lambda_m \approx 5\text{--}6 \times \text{FWHM}$ (Fischera & Martin 2012; $= 4 \times$ the flat diameter, Inutsuka & Miyama 1992) and the dotted line the classical *critical* wavelength $\lambda_{\text{cr}} \approx 2.6 \times \text{FWHM}$; every region lies well below both. The grey band marks the FilFinder skeleton range ($\approx 0.4\text{--}0.8$), and the vertical grey bars give the per-region min–max convention envelope, spanning the width conventions and the skeleton-algorithm factor. These are descriptive envelopes rather than confidence intervals (Section 2.13).

spacing is a factor 1.3–2.2 larger, but on heavily bridged skeletons it is inflated by arclength jumps across branches, so we treat it as an upper descriptor. (iii) The length per core is $L/N = 0.34\text{--}0.58\text{ pc}$, that is $S_{\text{global}} \approx 3.4$ (Orion B), 3.4 (Aquila), 4.5 (Taurus) and 5.8 (Perseus), a factor ≈ 2 larger than the median adjacent spacing in every cloud.

Two contributions produce that factor of two, and neither is small-scale clustering. The median-to-mean rise ($\approx 40\text{--}120$ per cent) is the generic skew of a broad gap distribution, a pure exponential already giving 1.44; the further rise to L/N is dilution by extensive core-free filament. The line density restricted to core-bearing stretches is intermediate and still mildly sub-classical, so the near-classical S_{global} is not evidence of a uniform classical spacing.

A *local-width* normalisation supports the same reading. Fitting the beam-deconvolved transverse width at the midpoint of every adjacent-core interval (146–545 reliable fits per cloud) gives median $\Delta s/W_{\text{local}} = 0.61, 0.91, 1.05$ and 0.99 (Taurus, Orion B, Perseus, Aquila), all sub-classical, with only a weak spacing–width correlation ($r = -0.06$ to 0.31). These median-of-ratios values are if anything more sub-classical than the corresponding ratio-of-medians, so the per-segment normalisation strengthens the conclusion.

2.9 An exploratory diagnostic: the column-density-selected eligible length

Identifying L/N with the quantity a one-fragment-per-wavelength theory predicts assumes that the whole skeleton could have fragmented. Much of it could not: a subcritical stretch has no axial instability to express. The estimate below is exploratory: the criterion is a column-density proxy for a line mass, and it is partly determined by the cores it is used to interpret. None of the conclusions of this paper rests on it.

We define eligibility *independently of where the cores are*, using column density. A filament of width 0.10 pc reaches the

critical line mass $M_{\text{line,crit}} = 2c_s^2/G \simeq 16 M_{\odot} \text{ pc}^{-1}$ at $N_{\text{H}_2} \simeq 7.2 \times 10^{21} \text{ cm}^{-2}$ ($\mu = 2.8$), and we take the *column-density-selected* eligible length L_{elig} to be the skeleton length above that threshold. This is a proxy, not a line-mass measurement: the conversion depends on width, inclination, temperature, background subtraction and both non-thermal and magnetic support. The eligible fractions differ greatly between clouds, 0.19, 0.91, 0.13 and 0.07 for Orion B, Aquila, Perseus and Taurus, so Orion B in particular is dominated by filament that could not have fragmented at all.

The criterion is partly endogenous. A core raises the column density of its own surroundings, so a threshold applied to the raw map can be crossed *by the core it is meant to be independent of*, selecting short eligible patches centred on cores. Between 25 and 34 per cent of skeleton pixels lie within one association half-width of a core, so this is not a small effect. We therefore rebuilt the criterion on a core-masked background: pixels within a masking radius of an associated core have their column density replaced by interpolation along the spine from untainted pixels, and the threshold is applied to that reconstruction.

The correction is substantial. On the core-masked background S_{elig} becomes 2.0, 3.2, 4.1 and 3.1, against the raw 1.6, 3.2, 2.0 and 0.8. Aquila is unchanged because almost all of its skeleton is eligible either way, but Perseus doubles and Taurus rises by nearly a factor of four, and N_{elig} falls from 199 to 57 in Perseus and from 186 to 13 in Taurus.

The statistic is also sensitive to the masking radius itself (Appendix E). Only Orion B and Aquila are comparatively stable against it and retain more than 230 eligible cores; Perseus drifts over 3.6–4.5 with a bootstrap uncertainty of $^{+1.1}_{-0.9}$ on 57 cores, and Taurus does not stabilise at any radius.

Sensitivity to the threshold itself is milder and runs the other way: raising it from 4×10^{21} to $1.2 \times 10^{22} \text{ cm}^{-2}$ moves S_{elig} monotonically downwards, so the fiducial choice is conservative, and adopting each region’s own Plummer width

or $T = 15$ K changes it by less than 20 per cent. A virial threshold, $M_{\text{line,vir}} = 2(c_s^2 + \sigma_{\text{nt}}^2)/G$, lowers it further, but treating σ_{nt} as isotropic support is a prescription rather than a measurement and beyond the transonic case the statistic degenerates.

What distinguishes a fertile stretch from a sterile one? Supercriticality is necessary but not sufficient: in Aquila 90 per cent of the crest is eligible yet only 24 per cent carries a core. Local compression, longitudinal velocity gradients of $\sim 1 \text{ km s}^{-1} \text{ pc}^{-1}$ (Kirk et al. 2013; Hacar et al. 2013), and spatially varying non-thermal support are all plausible controls, and none can be tested with column density alone. Deciding among them requires resolved kinematics per stretch, so the occupancy is reported here as a measured property whose cause is undetermined.

The message is limited but useful: a physically motivated restriction to nominally unstable crest moves S_{local} and S_{global} towards each other, but current column-density data do not permit a sufficiently independent determination of the unstable filament length. Resolved kinematics per stretch would be needed to convert this diagnostic into a measurement, and none of the conclusions below depends on it.

2.10 Does the local/global contrast survive changing the definition of a filament?

A skeleton defines L itself, so changing the extraction algorithm alters the spacing, the length, the core association and the topology together. DisPerSE and FilFinder are not two noisy estimates of one observable but partly different definitions of the object, and we therefore recomputed *all* the principal statistics under both (Table 6).

The FilFinder networks are shorter (67–121 pc against 106–337 pc) and associate far fewer catalogue cores, so both statistics move, but they move in *opposite* directions: S_{local} becomes shorter still (0.54–1.54 against 1.66–2.97) while S_{global} becomes much larger (9.1–16.7 against 3.4–5.8). The local/global contrast therefore survives the change of ontology and in fact widens, from a factor ≈ 2 to ≈ 6 –17, and the occupancy fraction stays low under both. We regard this as one of the more secure results here, precisely because the two algorithms disagree so strongly about what a filament is.

At a high-column hub, FilFinder recovers the main ridge but adds long, almost straight spokes that leave the emission entirely (Fig. 1). These are most likely medial-axis artefacts of a large connected mask rather than resolved sub-filaments, though we have not quantified what fraction of the total length behaves this way and column density alone cannot decide: position–position–velocity data are needed to establish which network corresponds to a coherent physical object. We therefore base our quantitative conclusions on the persistent-crest network and carry FilFinder as an ontology test.

Coverage bias. Between 41 and 86 per cent of catalogue cores associate with the skeleton (Table 10), and the rejected cores have projected separations 1.5–3.0 times *larger*, so coverage selection biases the adjacent spacing short. Relaxing the association half-width from 0.06 to 0.30 pc raises coverage to 70–97 per cent and lengthens the median adjacent spacing by only 3–16 per cent (Fig. 3c); S_{local} plateaus and the bias is bounded at $\lesssim 16$ per cent.

Table 6. Principal statistics under both filament definitions, at a 0.06 pc association half-width and a 0.06 pc occupancy half-width. $F(1 \text{ pc})$ is omitted where the FilFinder components are mostly shorter than the cell. The local and global statistics move in opposite senses between the two algorithms, so their contrast is not an artefact of either ontology.

Region	Skeleton	L (pc)	N_{assoc}	$(L/N)/W$	$s_{\text{ACS,med}}/W$	f_{occ}
Orion B	DisPerSE	337	995	3.4	1.77	0.27
	FilFinder	121	73	16.6	1.54	0.06
Aquila	DisPerSE	106	308	3.4	1.66	0.27
	FilFinder	67	40	16.7	0.97	0.05
Perseus	DisPerSE	290	501	5.8	2.97	0.12
	FilFinder	78	85	9.1	1.05	0.09
Taurus	DisPerSE	209	463	4.5	1.81	0.06
	FilFinder	111	120	9.3	0.54	0.05

The unconditional statistic behaves differently, and this is a genuine limitation. Because adding cores at fixed L necessarily raises the line density, S_{global} falls monotonically with association radius, from 3.6 to 2.1 in Orion B, 3.6 to 2.0 in Aquila, 6.6 to 4.2 in Perseus and 5.1 to 4.0 in Taurus. The near-classical value is robust in Perseus and Taurus but in Orion B and Aquila holds only near the adopted radius. The local/global contrast persists in all four clouds; its size in the two low-coverage clouds depends on how generously cores are assigned.

2.11 Spatial inhomogeneity: filament occupancy and scale-dependent over-dispersion

The gap between S_{local} and S_{global} implies that cores occupy preferential stretches. This can be measured. We assign each associated core an occupancy interval of ± 0.06 pc along its spine, merge overlapping intervals, and define the occupancy fraction f_{occ} as the fraction of skeleton length that is core-bearing; we also count cores in 1 pc cells and form the Fano factor $F = \sigma^2/\mu$ and the Gini coefficient of those counts (Table 7, Fig. 4).

Between 6 and 27 per cent of the skeleton carries cores, so most filament length is empty at HGBS sensitivity. The Fano factor exceeds unity everywhere (1.6, 2.8, 5.7, 25.2 for Aquila, Orion B, Perseus and Taurus) and the Gini coefficients are 0.36–0.57. In Orion B, 92 pc of the 337 pc skeleton is occupied, a further 155 pc lies in core-free runs inside core-bearing components, and roughly 90 pc is in components with no catalogued core at all.

Because f_{occ} necessarily grows with the assumed interval, we report it against ℓ (Fig. 3a). Taurus rises only from 0.052 to 0.078 and Perseus from 0.078 to 0.202 as ℓ increases tenfold, so widening the interval adds almost nothing; Orion B and Aquila climb to ≈ 0.6 and are much less segregated. The Fano factor is the more robust statistic because it needs no occupancy convention: $F(\ell) > 1$ across $\ell = 0.25$ –4 pc in Taurus (9.4–25), Perseus (2.3–8.4) and Orion B (1.2–4.9), rising monotonically with ℓ . Aquila is the exception, consistent with Poisson below 1 pc.

Is this more than the filament’s own structure implies? A Fano factor above unity compares the cores with a *homogeneous* Poisson process, and a filament is not homogeneous. We therefore built an inhomogeneous null whose in-

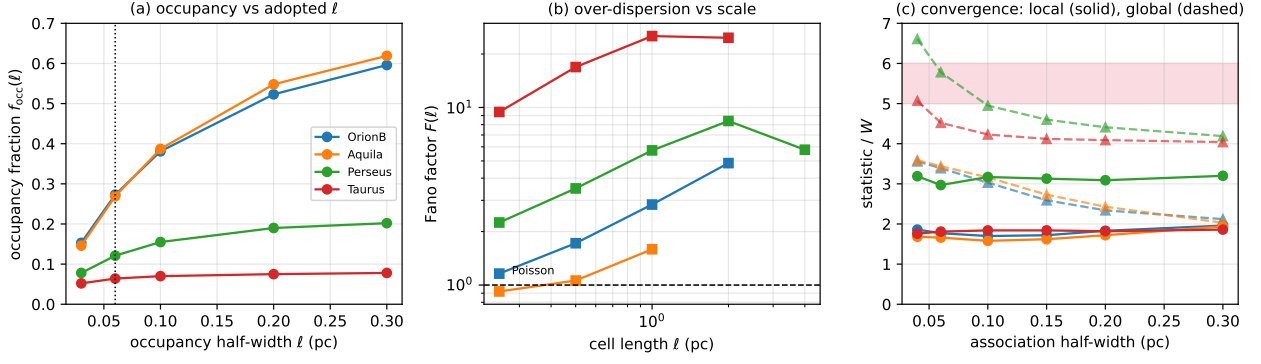


Figure 3. Scale and convention tests of the inhomogeneity result. (a) Occupancy fraction against the adopted half-width ℓ (dotted line: the 0.06 pc value quoted in the text). Taurus and Perseus saturate, so their low occupancy is not a consequence of the chosen ℓ ; Orion B and Aquila rise steadily and are less strongly segregated. (b) Fano factor against cell length. Over-dispersion ($F > 1$, dashed) persists across 0.25–4 pc in Taurus, Perseus and Orion B, but Aquila is consistent with Poisson below 1 pc. (c) Convergence against association half-width: the conditional $s_{\text{ACS,med}}/W$ (solid) plateaus, whereas the unconditional $(L/N)/W$ (dashed) falls as more cores are admitted, approaching the classical band (shaded) only in Perseus and Taurus.

tensity is conditioned on the local column density, measured $\lambda(N_{\text{H}_2})$ from the data, scattered synthetic cores along the actual skeletons and passed them through the same pipeline, in 10^4 realisations per cloud (Appendix F). In Orion B, Aquila and Perseus the observed F lies *below* the null median ($p = 0.98, 0.88, 0.96$): the column-density structure alone accounts for all of the over-dispersion. Only Taurus exceeds its null ($F = 25.2$ against a median of 13.4; $p = 2 \times 10^{-4}$). Refitting the intensity out of sample, from the other three clouds, leaves this unchanged.

One bias remains and it works against us: cores raise the column density of their own surroundings, so the pixels to which the null assigns the highest intensity are the core sites themselves, and the null is too easily satisfied. Masking a 0.06 pc radius around each associated core changes the intensity in the highest column bins by factors of 4–10. The three non-detections are therefore conservative rather than secure, and the Taurus excess is correspondingly strengthened. Excess clumping is therefore detected in one of the four clouds and not detected in the other three, which is not the same as its being absent there.

Two length scales coexist. On sub-parsec scales the gap distribution shows a deficit of the smallest separations (Section 2.12); on parsec scales the counts are strongly over-dispersed. Cores form in groups with short internal separations, and the groups are separated by long stretches that have produced none. This accounts for the factor of about two between S_{local} and S_{global} without appealing to small-scale clustering, and it is the central observational result of this paper: what a fragmentation model must reproduce is a spatial point process, not a single spacing.

2.12 Regularity of the core spacing

Classical fragmentation predicts a quasi-periodic chain; fibre projection or a turbulent cascade predicts a broad distribution with no comb. Ordering associated cores by arclength and measuring adjacent gaps gives a coefficient of variation of order unity (Taurus 1.36, Orion B 0.83, Perseus 0.96, Aquila 0.65; Fig. 6).

Table 7. Spatial-inhomogeneity diagnostics. L_{tot} is the total skeleton length, L_{core} the core-bearing length (union of ± 0.06 pc intervals about each associated core), and f_{occ} is their ratio. f_{assoc} is the fraction of catalogue cores associated with the skeleton, F the Fano factor of counts in 1 pc cells and G their Gini coefficient. Both association half-widths are listed.

Region	half (pc)	L_{tot} (pc)	L_{core} (pc)	f_{occ}	f_{assoc}	L/N W	L_{core}/N W	F	G
Orion B	0.06	337	92	0.27	0.53	3.4	0.9	2.8	0.49
	0.30	337	117	0.35	0.85	2.1	0.7	4.5	0.50
Aquila	0.06	106	29	0.27	0.41	3.4	0.9	1.6	0.40
	0.30	106	38	0.36	0.70	2.0	0.7	2.1	0.36
Perseus	0.06	290	35	0.12	0.61	5.8	0.7	5.7	0.43
	0.30	290	40	0.14	0.85	4.2	0.6	10.5	0.47
Taurus	0.06	210	13	0.06	0.86	4.5	0.3	25.2	0.57
	0.30	210	14	0.07	0.97	4.0	0.3	27.3	0.57

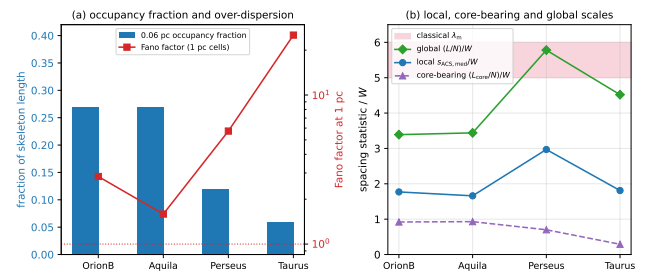


Figure 4. Spatial inhomogeneity of core occurrence along the filaments. (a) The 0.06 pc occupancy fraction (bars, left axis) is 6–27 per cent, while the Fano factor of counts in 1 pc cells (red, right axis, logarithmic) exceeds the Poisson value of unity in every cloud. (b) The three normalised spacing statistics: the local adjacent-core spacing is well below the classical band, the global $(L/N)/W$ approaches it, and the core-bearing $(L_{\text{core}}/N)/W$ lies below both because the occupied stretches are densely populated. The separation between the blue and green curves measures how unevenly the cores occupy the crest.

Goodness-of-fit tests were calibrated by parametric bootstrap ($N_{\text{boot}} = 10^5$), resampling from the fitted distribution to build the exact null distribution of the Kolmogorov–Smirnov statistic. The result is two-sided: the adopted periodic-comb-plus-25-per-cent-jitter model is rejected ($p_{\text{boot}} < 10^{-5}$ in every cloud) *and* so is the pure exponential ($p_{\text{boot}} < 10^{-5}$ in three clouds, 1.7×10^{-3} in Perseus). Comparing five renewal models by the corrected Akaike criterion makes this quantitative: $\Delta\text{AICc} = 37\text{--}246$ against Poisson and $81\text{--}742$ against the periodic comb in every cloud. The specific hard-core (shifted-exponential) model is not uniquely preferred once broader alternatives are admitted, a positively-skewed lognormal fitting best in three clouds, so the data favour a broad, skewed distribution truncated at small separations rather than a sharp physical hard core.

The clouds fail the exponential for different reasons and should not be summarised together (Table 8). Writing R for the observed median gap divided by that expected for an exponential of the same mean, only Orion B (1.03) and Aquila (1.14) show a deficit of the smallest separations; Perseus (0.92) and Taurus (0.64) show an excess.

The small-separation deficit. The gap distributions are truncated below 0.011–0.044 pc. This range lies entirely inside the extraction floor: the $18.2''$ beam corresponds to 0.012–0.039 pc across our distance range and GETSOURCES does not reliably separate sources closer than about one beam. We therefore treat the truncation as observational and attach no physical interpretation to it. Deciding whether any physical suppression is also present requires a completeness function $C(s, F_1/F_2, N_{\text{bg}})$ measured by injecting synthetic source pairs of varying separation, flux ratio, size and local background into the maps and re-extracting them through the HGBS pipeline, so that the gap distribution can be forward-modelled rather than simply truncated. That experiment is not attempted here.

Gap independence. A runs test about the median gap shows significant non-independence in Perseus ($z = -4.2$) and Taurus ($z = -5.4$), the same core-poor/core-rich alternation documented in Section 2.11. We therefore recomputed ΔAICc with a moving-block bootstrap that resamples contiguous runs of five adjacent gaps drawn from within single components (400 resamples per cloud). The effective sample size falls to 122 of 440 gaps in Taurus and 344 of 438 in Perseus, but the rejections survive: the 5th percentile of the bootstrap ΔAICc is 415, 45, 249 and 587 against the periodic model and 190, 71, 31 and 55 against Poisson. What does not survive is discrimination among the broad models, the shifted-exponential lying inside the lognormal’s bootstrap interval in three clouds.

No classical-scale periodicity is detected. The along-skeleton pair-correlation $g(s)$ (Fig. 5) shows no feature at $5\text{--}6W$. To decide whether that absence is significant we generated a 95 per cent envelope from the column-density-conditioned null of Appendix F: at the classical scale the observed g lies *inside* the envelope in every cloud (Orion B 1.02 in $[0.84, 1.26]$; Aquila 0.89 in $[0.58, 1.52]$; Perseus 1.02 in $[0.91, 1.14]$; Taurus 1.00 in $[0.91, 1.10]$). This is a non-detection at the 95 per cent level, not a demonstration of absence, and it is more informative than a comparison of me-



Figure 5. Along-skeleton pair-correlation function $g(s)$ for the four primary clouds (uniform-random expectation on the matched components divided out; $g = 1$ is no correlation). Dotted vertical lines mark the classical scale $5.5 W_{\text{RI}}$ per region. No cloud shows a peak at the classical scale ($g \approx 1$ there, slightly below unity in Orion B and Aquila); the only departures are a mild small-separation excess (Taurus) and weak region-envelope features near 1 pc. The classical periodic comb is therefore absent, consistent with a non-periodic, spatially inhomogeneous distribution bearing an observational small-separation deficit (Section 2.12).

Table 8. Cloud-by-cloud summary of the point-process diagnostics, to avoid summarising four different behaviours as one. R is the observed median gap divided by the median expected for an exponential of the same mean ($R > 1$ deficit of small gaps, $R < 1$ excess); $F(1\text{ pc})$ is the Fano factor; “null” compares F with the inhomogeneous, column-density-conditioned null of Section 2.12. The clouds do not behave alike, and only Taurus exceeds its inhomogeneous null. With $N = 4$ this is a single case, not a demonstrated population split; and because the null is built partly from column density the cores themselves raise, the three non-detections are conservative.

Region	CoV	R	$F(1\text{ pc})$	vs null	preferred gap model
Orion B	0.83	1.03	2.8	below ($p = 0.99$)	lognormal
Aquila	0.65	1.14	1.6	below ($p = 0.95$)	lognormal
Perseus	0.96	0.92	5.7	below ($p = 0.97$)	shifted-exp.
Taurus	1.36	0.64	25.2	above ($p = 0.005$)	lognormal

dian spacings because it does not compress the point process into a single number.

2.13 Systematic error budget

Not all of the checks reported above carry equal weight, and it is worth saying which are decisive. With $N = 4$ clouds, the tests that bear directly on the conclusions are those that could change the *sign* of a result: the skeleton-ontology comparison (Section 2.10), the coverage and association-radius sweeps, the core-population splits (Section 2.15) and the column-density-conditioned null (Section 2.11). The remainder, including the width and persistence conventions, the injection–recovery corrections and the eligible-length sweeps, are illustrative: they move numbers within the envelope already quoted and none of them reverses a conclusion. Table 9 collects the terms. We separate *statistical/sensitivity* terms, which would change under resampling, from *sys-*

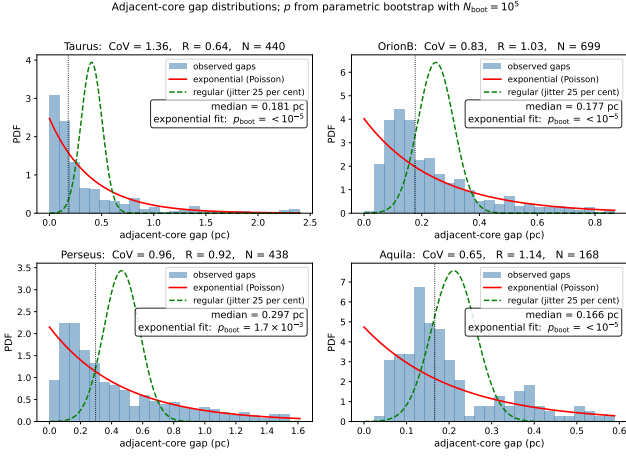


Figure 6. Adjacent-core gap distributions along the skeletons of the four primary regions (histograms), with exponential/Poisson (solid) and regular/periodic (dashed) reference curves; each panel gives the coefficient of variation CoV, the 1-D Clark-Evans ratio R , and the exponential goodness-of-fit p -value calibrated by parametric bootstrap (p_{boot} ; the null distribution of the KS statistic is built by resampling from the fitted exponential, removing the estimated-parameter bias of the standard test). The result is two-sided: the regular/periodic model is rejected ($p_{\text{boot}} < 10^{-5}$), the parametric-bootstrap floor, $1/N_{\text{boot}}$ with $N_{\text{boot}} = 10^5$ in every region, and the pure-exponential (Poisson) model is also rejected ($p_{\text{boot}} < 10^{-5}$ (Perseus 1.7×10^{-3}) in all four), so in each of the four clouds the cores are neither periodic nor pure-Poisson; consistent with a point process that is limited at small separations by the extraction resolution but over-dispersed on parsec scales (the deficit of small gaps, $R > 1$; largely instrumental).

tematic offsets and from modelling terms arising in the simulations. Uncorrelated statistical terms are combined in quadrature, $\sigma_{\text{sys}}^2 = \sum_i \sigma_i^2$; the multiplicative width- and skeleton-convention factors are applied afterwards as separate analysis-choice envelopes rather than added into that sum. The adopted $(s_{\text{ACS,med}}/W)_0 \approx 1.4$ carries residual pipeline bias (~ 10 per cent), protostellar migration (~ 7.5 per cent), between-region scatter (~ 4 per cent) and a $\lesssim 5$ per cent manual-editing proxy in quadrature, with the width convention (~ 25 per cent) and the skeleton algorithm (a factor ~ 2) applied on top. Because the dominant term is a deterministic offset we quote a min-max envelope, $S_{\text{local}} \approx 0.8\text{--}1.9$ for DisPerSE, rather than a Gaussian σ .

Projection acts against our conclusion and is shown in Table 4 for that reason. For an isotropic orientation distribution $s_{\text{obs}}/s_{\text{true}} = \sin i$ has median 0.87, so deprojection raises the intrinsic spacing by 15–27 per cent, moving the adopted 1.4 to 1.6–1.8. We do not apply it, because the orientation distribution within a single cloud need not be isotropic, but we carry it as a one-sided systematic.

Skeleton-algorithm dependence is the largest observational term. The DisPerSE and FilFinder values (≈ 1.9 and $\approx 0.4\text{--}0.8$) bracket a genuine ambiguity in what constitutes “the filament” rather than a measurement error, and a matched-threshold FilFinder extraction shows the inferred spacing is itself strongly sensitive to the mask threshold, so the factor of two is if anything a lower bound. We base our quantitative conclusions on the persistent-crest network, which is

Table 9. Summary of the terms on the λ/W comparison, grouped as *observational* (on $s_{\text{ACS,med}}$), *simulation* (on λ_{turn}) and *comparison* (terms that enter only when the two are matched). Statistical/sensitivity terms are combined in quadrature; the width- and skeleton-convention factors are applied afterwards as a separate multiplicative *analysis-choice envelope*, not as Gaussian uncertainties.

Source	Type	Magnitude
<i>Observational (on $s_{\text{ACS,med}}$)</i>		
Bootstrap / cloud-to-cloud	sensitivity	$\sigma \approx 0.4$ ($\sim 20\%$)
ACS pipeline bias	systematic	$\sim 10\%$ (corrected)
Protostellar migration	sensitivity	$\sim 7.5\%$
Projection (deprojection)	systematic (one-sided, raises s)	$+15\text{--}27\%$
LOS blending (Aquila)	systematic (one-sided, shortens s)	$\lesssim 50\%$ worst case
Skeleton-coverage selection	systematic (one-sided, shortens s)	$\lesssim 16\%$ (bounded)
Width convention, fixed vs region	analysis-choice env.	$\sim 25\%$
Skeleton algorithm, DisPerSE/FilFinder	analysis-choice env.	factor ~ 2
Manual skeleton editing	systematic (proxy)	$\lesssim 5\%$
<i>Simulation (on λ_{turn})</i>		
Growth-rate resolution/epoch	modelling	$\lesssim 5\%$ (converged)
Turbulence sensitivity	sensitivity	$< 5\%$ (single realisation)
Line-mass (f) dependence	modelling	$\sim 25\%$
<i>Comparison (matching the two)</i>		
Gaussian vs Plummer FWHM	systematic	factor ~ 1.5
T_1 width normalisation (η_W)	systematic	$\sim 30\%$

the physically better-motivated definition of a filament ridge, and treat FilFinder as an ontology test rather than an equally weighted measurement.

2.14 Pipeline counts, coverage and population splits

The adjacent-core measurement thresholds the deposited DisPerSE skeleton, bridges fragmented segments by morphological closing and re-skeletonisation, associates each core to the nearest spine within 0.06 pc, orders the associated cores along each component by graph arclength (double-BFS diameter) and takes adjacent separations. Table 10 lists the resulting counts.

The associated fraction ranges from 0.41 to 0.86, so in Orion B and Aquila the spacing is measured on roughly half the catalogue. This selection is not neutral, but it acts *towards* the sub-classical result: the unassociated cores lie on fainter segments and have projected separations 1.5–3.0 times larger, and relaxing the association half-width to 0.30 pc raises the coverage to 70–97 per cent while lengthening the median adjacent spacing by only 3–16 per cent (Section 2.10). The bias on S_{local} is therefore one-sided and bounded at $\lesssim 16$

Table 10. Measurement counts per primary region: catalogue cores, cores associated to a skeleton within 0.06 pc, the associated fraction, and the number of adjacent-core pairs entering the median.

Region	catalogue	associated	f_{assoc}	ACS pairs
Orion B	1870	995	0.53	699
Aquila	749	308	0.41	168
Perseus	816	501	0.61	438
Taurus	536	463	0.86	440

per cent. Sweeping the half-width over 0.04–0.08 pc changes the per-region median by 4–7 per cent, so the result is not sensitive to that choice.

Restricting to the two clouds with the highest coverage, Perseus and Taurus, gives the same median as the full four-region sample. The dependence on the core population is examined in Section 2.15.

2.15 Core population splits

The HGBS catalogues contain unbound starless cores, prestellar cores and protostellar sources. These are not equivalent for the present purpose: protostellar objects have had time to accrete, migrate and interact, while unbound starless cores need not be fragmentation products at all. We therefore repeated the principal statistics on three nested samples per cloud (Table 11).

The conditional statistic is essentially invariant. S_{local} changes by at most 0.2 between the all-core and prestellar-only samples in every cloud, and the coefficient of variation of the gaps is unchanged to within 0.1. The unconditional statistic behaves in the opposite way, rising steeply as the sample is restricted, from 3.4 to 5.9 in Orion B, 3.4 to 4.8 in Aquila and 5.8 to 10.3 in Perseus, simply because L is fixed while N falls.

That combination sharpens rather than weakens the central result. The separation between the local and global statistics is not produced by the inclusion of evolved or unbound objects; restricting to the cores most likely to be genuine fragmentation products *widens* it, and pushes S_{global} from near-classical to super-classical in three of the four clouds. It also shows that S_{global} is sensitive to catalogue selection in a way that S_{local} is not, which is a further reason to treat the comparison of S_{global} with λ_{m} as a diagnostic rather than a measurement. The Taurus prestellar sample contains only 54 cores and its S_{global} is not meaningful; we quote it only for completeness.

2.16 Comparison with the literature

Our revised-distance ACS measurements (Table 12) differ from the original HGBS values by the region-dependent distance revisions; the core-count-weighted primary-region mean rises only ~ 10 per cent, dominated by Aquila. Toggling each methodological choice on the Orion B data at fixed distance shows that the *statistic definition* dominates: switching from ACS to the pairwise median inflates the spacing from 0.207 to 0.313 pc and accounts for essentially the whole apparent shift, while skeleton re-extraction, association half-width, arclength ordering and the distance revision are each

Table 11. Principal statistics for three nested core samples. “All” is the full published catalogue; “starless+prestellar” removes protostellar sources; “prestellar” retains only cores classified as prestellar. N_{assoc} is the number associated with a skeleton, CoV the coefficient of variation of the adjacent gaps. The Taurus prestellar row rests on 53 associated cores and its S_{global} should not be interpreted.

Cloud	sample	N_{assoc}	S_{local}	S_{global}	CoV
Orion B	all	995	1.77	3.4	0.83
	starless+prestellar	961	1.79	3.5	0.83
	prestellar	569	1.83	5.9	0.82
Aquila	all	308	1.66	3.4	0.65
	starless+prestellar	267	1.78	4.0	0.62
	prestellar	223	1.78	4.8	0.73
Perseus	all	501	2.97	5.8	0.96
	starless+prestellar	438	2.99	6.6	0.95
	prestellar	282	2.83	10.3	1.01
Taurus	all	463	1.81	4.5	1.36
	starless+prestellar	424	1.84	4.9	1.34
	prestellar	53	2.04	(39.5)	1.30

≤ 15 per cent and largely cancel. On our primary statistic Orion B agrees with Könyves et al. (2020) to within 3 per cent.

The published values we could check are local or along-fibre separations, not all-pairs medians, which is why they agree with our ACS column rather than our pairwise column. The region-extent bias applies wherever an all-pairs or pooled statistic is adopted, including the comparability statistic carried here and in cross-region compilations, and we make no claim that it affects these particular published values.

2.17 What constitutes the theoretical filament?

The factor-of-two DisPerSE-versus-FilFinder difference (Fig. 1) is not measurement noise: the two algorithms *define different objects* (persistent crests versus the medial axis of a masked region), so there is no single, algorithm-independent “filament” whose spacing they should reproduce. Our robustness claim applies to the *sign* of the deficit, every convention giving a sub-classical S_{local} , while its *magnitude* cannot be assigned a unique value: the same data yield $S_{\text{local}} \approx 1.0$ (region-Plummer FilFinder), ≈ 1.4 (region-Plummer DisPerSE) or ≈ 1.9 (fixed-width DisPerSE). Because segmentation alone moves the answer by as much as the differences between the physical models compared in Section 5, it should not be read as discriminating between them. Figure 7 makes the resulting envelope explicit.

There is a more fundamental point behind this. The classical calculation is an eigenmode of a single, coherent, approximately cylindrical, self-gravitating object. Neither a persistent column-density crest nor the medial axis of a mask guarantees that such an object exists along the traced path: both are two-dimensional constructions on a projected scalar field, and either may connect material that is unrelated in velocity, or divide material that is dynamically single. Before any skeleton-derived length is compared with a cylinder eigenmode, the segment must be shown to correspond to a velocity-coherent structure. That cannot be established from column density, and it is the reason the comparison in

Table 12. Comparison with published HGBS spacing measurements. Perseus and Taurus are the two regions in which our own gap sequence shows significant non-independence (runs test; Section 2.12) and in which the post-masking eligible-core samples collapse (Section 2.9); their entries should be compared with the published values with those caveats in mind.

Region	ACS (pc)	Orig. HGBS	pairwise (pc)	Lit. (pc)	Lit. statistic	Reference
Primary sample						
Orion B	0.207	0.212	0.313	0.20–0.30	local	Könyves et al. (2020)
Aquila	0.210	0.206	0.346	0.22–0.26	local	Könyves et al. (2015)
Perseus	0.263	0.199	0.248	0.22	local	Pezzuto et al. (2021)
Taurus	0.124	0.206	0.198	0.19	along-fibre	Hacar et al. (2013)
Limited sample						
Ophiuchus	,	0.197	0.206	0.18	local	Könyves et al. (2020)
Serpens	,	0.188	0.331	0.17–0.19	local	Könyves et al. (2020)
TMC1	,	0.203	0.195	~0.20	along-fibre	Hacar et al. (2013)
CRA	,	0.212	0.248	~0.21	local	Könyves et al. (2020)

Notes ^a The “Lit. statistic” column records what kind of statistic each cited value represents. All are *local* separation measures, either nearest-neighbour or along-fibre adjacent separations, rather than all-pairs medians, and that is precisely why they agree with our ACS column rather than with our pairwise column; the exact definitions differ in detail between papers. The region-extent bias identified here therefore applies wherever an all-pairs or pooled separation statistic is adopted, including the comparability statistic carried in this paper and in cross-region compilations, and not to these particular published values. ^b The primary statistic is the *adjacent-core* (ACS) spacing (bold); the “pairwise” column is a legacy comparator only, inflated towards $L/3$ (Section 2.6) and not a physical spacing. ACS spacings for the limited sample are omitted. Perseus is the one region where the ACS (0.263) slightly exceeds the pairwise median, because arclength ordering on its heavily bridged skeleton inflates path distances; we verified this is not a bridging artefact, an independent minimum-spanning-tree ordering of the same cores gives a consistent value, insensitive to the morphological bridging, and both remain sub-classical. ^c “Original HGBS” values use pre-Gaia distances; literature ranges reflect different methods and distance assumptions. ^d For Orion B the pre-Gaia distance was already ~ 400 pc (Könyves et al. 2020), so the pairwise $0.212 \rightarrow 0.313$ pc difference is a statistic-definition effect, not a distance revision.

this paper is presented as a diagnostic rather than a test, and the reason the decisive follow-up is kinematic: dense-gas position–position–velocity data that identify which of the two networks, if either, traces a dynamically coherent cylinder.

3 THEORETICAL FRAMEWORK

3.1 Classical fragmentation theory

For an isothermal, self-gravitating cylinder of density ρ_0 and sound speed c_s , the critical line mass for gravitational instability is (Ostriker 1964):

$$\mu_{\text{crit}} = \frac{2c_s^2}{G} \quad (1)$$

The most unstable fragmentation wavelength is, as an *exact* result for the isothermal equilibrium cylinder (Inutsuka & Miyama 1992),

$$\lambda_{\text{IM92}} = 22 H \approx 4 \times (2R_{\text{flat}}), \quad (2)$$

where $H = c_s(4\pi G\rho_c)^{-1/2}$ is the thermal scale height, $R_{\text{flat}} = \sqrt{8}H$ the flattening radius, and $W_{\text{fil}} \approx 0.1$ pc the characteristic filament FWHM. Solving the dispersion relation exactly (Appendix C; Fischera & Martin 2012) gives a fastest-growing wavelength $\lambda_m \approx 5 \times W_{\text{fil}}$ near-critical, rising to $\approx 6 \times W_{\text{fil}}$ at low line mass, and a critical wavelength $\lambda_{\text{cr}} \approx 2.6 \times W_{\text{fil}}$. The often-quoted “ $\lambda \approx 4 \times$ the width” is $4 \times$ the flat *diameter* $2R_{\text{flat}}$ (Inutsuka & Miyama 1992), which equals $\approx 5\text{--}6 \times$ the *FWHM*; taking it as $4 \times$ the FWHM understates the classical scale. We therefore adopt $\lambda_m \approx 5\text{--}6 W_{\text{fil}}$ (Fischera & Martin 2012) as the benchmark. Because we compare in FWHM units and forward-model the simulations to a beam-convolved Plummer FWHM (Section 3.2),

Table 13. Where the $5\text{--}6 \times \text{FWHM}$ benchmark comes from, for the isothermal equilibrium (Ostriker, $p = 4$) cylinder. The ratios in the last column are exact for that equilibrium; the final row is the one that is *not* invariant when the observed profile has $p \simeq 2$.

Quantity	Definition	Value
H	$c_s(4\pi G\rho_c)^{-1/2}$, scale height	1
R_{flat}	flattening radius, $\sqrt{8}H$	$2.83 H$
$2R_{\text{flat}}$	flat diameter	$5.66 H$
W_{FWHM}	FWHM of the $p = 4$ profile	$\approx 3.9 H$
λ_{cr}	cutoff wavelength	$\approx 10 H (\approx 2.6 W)$
λ_m	fastest-growing wavelength, $22H$	$22 H (\approx 5.6 W)$
$\lambda_m/(2R_{\text{flat}})$	the “ $4 \times$ diameter” form	3.9
$\lambda_m/W_{\text{FWHM}}$	the benchmark used here	5–6

the sub-classical deficit does not depend on the equilibrium-vs-observed profile-index difference ($p = 4$ vs $p \simeq 2$), and we compare as the density-independent ratio λ/W_{fil} . The deficit is not new in sign: Fischera & Martin (2012) noted observed cores sit “about the FWHM” apart, and Zhang et al. (2020) find a spacing $\approx 1.3 \times$ the width in California.

Two things follow. The ratio $\lambda_m/H = 22$ and the conversion $\lambda_m \approx 4 \times (2R_{\text{flat}})$ are properties of the $p = 4$ equilibrium and are exact within it. What is *not* invariant is $\lambda_m/W_{\text{FWHM}}$, because the FWHM of a $p \simeq 2$ Plummer profile is a different fraction of H from that of a $p = 4$ profile, and beam convolution broadens it further. The observational benchmark is therefore convention-dependent at the tens-of-per-cent level even before any measurement error, which is one reason we treat the simulated comparison through forward modelling (Section 3.2) rather than through this ratio.

With $f = \mu_{\text{line}}/\mu_{\text{crit}}$ and $t_{\text{frag}} \propto (f-1)^{-1/2}$, marginally supercritical filaments fragment slowly. At magnetically supercritical field strength ($\beta \gtrsim 0.2$) all configurations fragment for

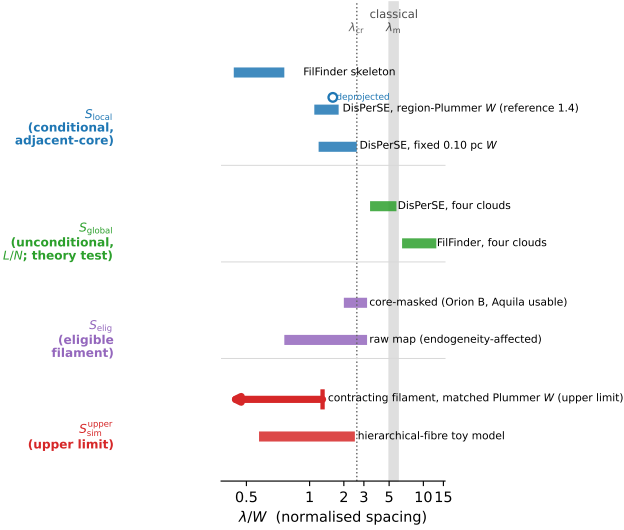


Figure 7. The full methodological envelope of λ/W , organised by which quantity is being measured rather than by convention alone. The four groups are the conditional adjacent-core spacing S_{local} (blue), the unconditional length per core $S_{\text{global}} = (L/N)/W$ (green), the same restricted to fragmentation-eligible filament S_{elig} (purple; the core-masked row is fiducial, the raw-map row shows the endogeneity-affected comparison), and the simulated upper bound $S_{\text{sim}}^{\text{upper}} = \lambda_{\text{turn}}/W$ together with the hierarchical-fibre toy model (red). Each bar spans the range across clouds and conventions; the grey band is the classical $\lambda_m \approx 5\text{--}6$ (Inutsuka & Miyama 1992; Fischera & Martin 2012) and the dotted line the critical $\lambda_{\text{cr}} \approx 2.6$. Grouping this way makes the central point of the paper visible directly: S_{local} lies well below the classical band under every convention, S_{global} reaches it, S_{elig} falls between the two, and $S_{\text{sim}}^{\text{upper}}$ is an *upper bound* that should not be read as a measured wavelength. Comparisons between groups are comparisons between different quantities and are only meaningful once that is stated (Section 2.17). The horizontal axis is symmetric-log below $\lambda/W = 1$. The $S_{\text{sim}}^{\text{upper}}$ bar is drawn with a left-pointing arrow because it is an upper limit, not a measured interval; the open marker on the S_{local} group shows where the adopted value moves if the isotropic deprojection factor of Table 4 is applied.

$\mathcal{M} \geq 1$ at every f tested once integrated adequately: above the critical line mass there is no *thermal* stability boundary. This is distinct from *magnetic* stabilisation, where a strong longitudinal field ($\beta \lesssim 0.15$) suppresses collapse even for $f > 1$ (the “strong-field, magnetically supported” regime; the deposited material); the thermal (f) and magnetic (mass-to-flux) criticalities are physically distinct.

We adopt a single reference $W_{\text{fil}} = 0.10\text{ pc}$ for comparability with prior HGBS work, though its universality is debated (Panopoulou et al. 2017, 2022); the directly-measured region-Plummer widths (0.11–0.16 pc, $\lambda/W \approx 1.4$; Section 2.4) show the sub-classical conclusion is robust to the convention, carried as the $\sim 25\%$ width systematic (Section 2.13).

3.2 Width normalisation: W_{core} to W_{fil}

Three widths enter the simulation–observation comparison and must be kept separate (Fig. 8; Table 14): the initial Gaussian half-width $W_{\text{core}} = 0.3 \lambda_J$; the formation FWHM $W_{\text{form}} = 0.683 \lambda_J$ of the evolved projected profile

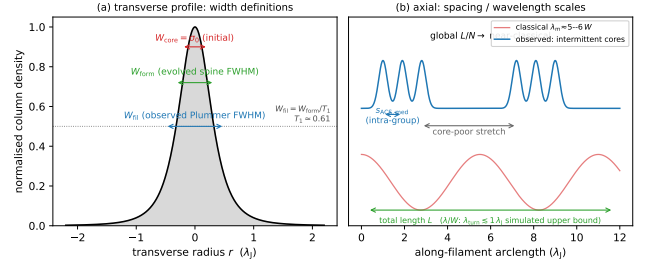


Figure 8. Schematic of the length-scale definitions used throughout (Table 1). (a) Transverse column-density profile, marking the initial half-width $W_{\text{core}} = \sigma_0$, the evolved spine FWHM W_{form} , and the observed beam-convolved Plummer FWHM $W_{\text{fil}} = W_{\text{form}}/T_1$ ($T_1 \approx 0.61$; Eq. 4). (b) Axial view contrasting the classical widely-spaced comb ($\lambda_m \approx 5\text{--}6 W$) with the observed unevenly-distributed cores: a short intra-group adjacent spacing $s_{\text{ACS,med}}$ within core-rich stretches separated by core-poor filament, whose filament-averaged line density L/N is near-classical, while the simulated fragmentation scale is an upper bound $\lambda_{\text{turn}} \lesssim 1 \lambda_J$.

($\approx 2.28 W_{\text{core}}$, slightly below the initial $2.355 W_{\text{core}}$ owing to contraction; resolution- and seed-converged to $< 1\%$); and the observed HGBS width W_{fil} , the beam-convolved Plummer FWHM (Arzoumanian et al. 2019), which we obtain by forward-modelling the simulated filaments through the full synthetic-HGBS pipeline (projection, beam convolution, Plummer fit). This gives $\eta_W = W_{\text{form}}/W_{\text{fil}} = 0.59\text{--}0.78$ and

$$T_1 \equiv \frac{W_{\text{form}}}{W_{\text{fil}}} = 0.61 \quad (\text{forward-model band } 0.59\text{--}0.78), \quad (3)$$

so that $W_{\text{fil}} = W_{\text{form}}/T_1 \approx 1.12 \lambda_J$. We quote the forward-model best-fit $T_1 = 0.61$; because it lies near the lower edge of the band, a mid-band $T_1 \approx 0.68$ would give a slightly *larger* (less sub-classical) λ/W_{fil} , so this normalisation is deliberately not the headline; the primary comparison uses the width-convention-independent growth-rate upper bound (Section 5), and this T_1 conversion is retained only as a cross-check. The observable is λ/W_{fil} ; because the fragmentation spacing λ is a physical length independent of the width definition, the conversion from the simulation-internal ratio λ/W_{core} (normalised by the initial $\sigma = 0.3 \lambda_J$) is

$$\left(\frac{\lambda}{W}\right)_{\text{fil}} = \frac{W_{\text{core}}}{W_{\text{fil}}} \left(\frac{\lambda}{W}\right)_{\text{core}} \approx 0.27 \left(\frac{\lambda}{W}\right)_{\text{core}}, \quad (4)$$

i.e. $\lambda/W_{\text{fil}} = (\lambda/\lambda_J)/1.12$; the prefactor is $0.27 = W_{\text{core}}/W_{\text{fil}} = 0.30/1.12$ (equivalently T_1 times the $\sigma \rightarrow \text{FWHM}$ factor, $0.61 \times 0.44 = 0.27$, using $T_1 W_{\text{core}}/W_{\text{form}}$), and using T_1 alone would overstate λ/W_{fil} by $\approx 2.3\times$. This prefactor is *not* a fixed constant: it carries the full $\approx 30\%$ T_1 band ($T_1 = 0.59\text{--}0.78$), so 0.27 is the best-fit value of a $\approx 0.26\text{--}0.34$ range. Because the observable is sensitive to how W_{fil} is defined, the primary comparison uses the width-convention-independent growth-rate wavelength (an upper limit; Section 5), and this normalisation is retained only as a cross-check.

For the ambient-confined supercritical ensemble ($\lambda/W_{\text{core}} = 3.27$) this gives $\lambda/W_{\text{fil}} \approx 0.9$.

Table 14. The three filament widths used in the simulation–observation comparison (all in Jeans lengths, $\lambda_J = 1$).

Symbol	Value (λ_J)	Definition
W_{core}	0.30	initial Gaussian half-width σ (input)
W_{form}	0.683	Gaussian FWHM of evolved projected profile, supercritical ensemble ($\simeq 2.28 W_{\text{core}}$)
W_{FWHM}	0.58	Gaussian FWHM at the growth-rate epoch, near-critical runs (narrower, already contracting); normalises Table 16
W_{fil}	≈ 1.12	beam-convolved Plummer FWHM ($= W_{\text{form}}/T_1$); the observable normalisation

3.3 Magnetic tension: a static-field estimate

For a filament threaded by a magnetic field at an angle θ to the axis (Alfvén speed v_A), a schematic oblique dispersion relation, valid only for a static background field, is (Nakamura et al. 1993):

$$\omega^2 = c_s^2 k^2 + v_A^2 k^2 \sin^2 \theta - 4\pi G \rho_0 F(kR) \quad (5)$$

where $F(kR)$ encodes the cylindrical geometry and θ is the field–axis angle. This is a *schematic static-field* relation that treats $\mathbf{B}$ as a fixed background and *cannot be used quantitatively* for a magnetised cylindrical equilibrium; in particular it does not describe a current-carrying (helical) equilibrium. We use it only as an analytic guide, so this paper contains *no quantitative model of magnetic tension*; it nonetheless fixes the *sign* of each idealised case, the magnetic term $v_A^2 k^2 \sin^2 \theta$ vanishing for a *longitudinal* field (reducing to the field-free thermal relation) and being maximal for a *perpendicular* one ($\theta = 90^\circ$, which lengthens the axial mode). Consistent with this, our longitudinal runs keep the growth-rate wavelength at order λ_J across $\beta = 0.3$ – 1.0 ($f = 1.1$; the $\beta = 1.0$ ladder gives $\lambda_{\text{turn}} = 1.14 \lambda_J$, Table 16) with no resolved β -trend, so the scale is set by the thermal Jeans length, not the field, and the shortening we measure is the dynamical/contraction shift, not magnetic tension. A genuinely shorter magnetic wavelength would require a force-balanced *helical*/toroidal field, whose stability is a cylindrical magnetostatic-equilibrium problem that Eq. (5) cannot address. The full case-by-case discussion of the longitudinal, perpendicular and helical geometries, and the sense in which the simulations *constrain only the specific initial field geometries simulated* rather than dispose of magnetic alternatives, is deferred to Section 6.3. The seeded helical run tested here is an *out-of-equilibrium pinch*, not a force-balanced equilibrium, and does *not* rule out the helical mechanism (Section 6.3).

3.4 Hierarchical fragmentation

Filament fragmentation is hierarchical: filaments fragment into velocity-coherent fibres, and fibres into cores (Hacar et al. 2013, 2018; Yang et al. 2024). Whether the fibre-to-core spacing is quasi-periodic at the classical scale is not established; the ALMA study of Yang et al. (2024) finds a clear filament–fibre–core hierarchy but only a weak quasi-periodic

Table 15. Schematic summary of the four magnetic-field configurations and their tested status.

Configuration	Tested?	Effect on axial λ
Uniform longitudinal	tested	unchanged (term vanishes)
Static perpendicular (resolved)	tested	suppresses axial fragmentation (spindle)
Seeded/dynamic helical	tested	pinches W ; does not shorten λ
<i>Not simulated in this work; listed for completeness</i>		
Force-balanced helical (Fiege–Pudritz)	not tested	could shorten by $\lesssim 2\times$ (order-of-magnitude inference from Fiege–Pudritz, not a result of this work; Section 6.3)

imprint at the core level. If HGBS filaments are bundles of fibres each fragmenting near the classical scale, measuring core spacings across the whole filament compresses the apparent spacing, in the observed direction. We test this quantitatively in Section 6.2, where a *toy geometrical model* of N -fibre bundles run through our ACS pipeline reproduces the observed $s_{\text{ACS,med}}/W$; it emerges as a viable but not unique explanation (single-filament dynamical fragmentation is also consistent), and confirming it requires fibre-resolved core-spacing analysis beyond the single Orion B region of Yang et al. (2024).

4 MHD SIMULATIONS

The calculations below test whether sub-equilibrium fragmentation scales are dynamically possible. They are not intended to reproduce the observed core point process, and no numerical value quoted in them should be read as a prediction of an observed spacing.

The calculations are exploratory. They are not a survey of filament fragmentation and they do not deliver a measured fragmentation wavelength: the growth-rate spectrum of a contracting filament has no resolved turnover, so every simulated number quoted here is an *upper bound* λ_{turn} on where a turnover could lie. Their role is to provide an existence proof, that fragmentation from a contracting rather than an equilibrium state is not constrained to the classical wavelength, calibrated by the equilibrium-recovery control of Section 5. They should be read as a counterexample to a theoretical expectation, not as a prediction to be matched number-for-number against the observations.

The full campaign comprises 2251 Athena++ runs across 18 grids, consuming of order 10^5 CPU-hours on Google Cloud E2 infrastructure; controlled numerical experiments, not models of individual HGBS filaments. The large majority are *exploratory* runs that map outcomes and timescales; the headline *wavelength* result rests instead on the load-bearing *production* runs (the 25-run near-critical growth-rate campaign, Table 16, and its 8-run 128^3 – 512^3 convergence ladder; Table 16), so the 2251-run total is *not* the statistical weight behind the wavelength. Two results emerge (Section 6.3): the *outcome* of a run (radial collapse or longitudinal beading) depends on regime and boundary conditions, so a single filament has no setup-independent fate, whereas the scale on

which it fragments is bounded above at a sub-classical value ($\approx 0.5\text{--}1\times$ the classical value; Section 5) for near-critical, longitudinal-field filaments (the perpendicular geometry is not numerically resolved). This section provides the evidence for both.

4.1 Simulation method

We performed self-gravitating simulations with Athena++ v21.0 (Stone et al. 2020), solving the standard ideal-MHD equations, continuity, momentum with the Lorentz force, ideal induction, and total energy, coupled to self-gravity through the Poisson equation $\nabla^2\phi = 4\pi G\rho$.

The production initial condition, stated in full. The production runs use a *Gaussian* filament embedded in a uniform ambient medium,

$$\rho(r) = \rho_{\text{bg}} + \rho_{\text{amp}} \exp\left(-\frac{r^2}{2W_{\text{core}}^2}\right), \quad \rho_{\text{amp}} = \frac{f M_{\text{line,crit}}}{2\pi W_{\text{core}}^2}, \quad (6)$$

with $r^2 = y^2 + z^2$, $\rho_{\text{bg}} = 1$ in code units, $M_{\text{line,crit}} = 2c_s^2/G$ and $W_{\text{core}} = 0.3\lambda_J$ the initial Gaussian half-width σ_0 of Table 1. The normalisation of ρ_{amp} is chosen so that $\int_0^\infty (\rho - \rho_{\text{bg}}) 2\pi r dr = f M_{\text{line,crit}}$ exactly; the line mass therefore converges without an imposed truncation radius, and f is imposed directly rather than inferred. There is no separate truncation: the filament merges into the uniform ambient medium, which also sets the confining pressure $P_{\text{bg}} = \rho_{\text{bg}} c_s^2$. The equilibrium-recovery control instead uses the Ostriker profile available in the same problem generator,

$$\rho(r) = \rho_{\text{bg}} + \frac{f \rho_{c,\text{ost}}}{[1 + (r/W_{\text{core}})^2]^2}, \quad \rho_{c,\text{ost}} = \frac{2c_s^2}{\pi G W_{\text{core}}^2}, \quad (7)$$

that is a Plummer profile of index $p = 4$, whose line mass also converges. Neither production nor control run uses a $p = 2$ profile; the observed filaments have $p \simeq 2$, which is why the width conversion of Section 3.2 is needed and why it is not invariant.

The remaining initial fields are $P(r) = \rho(r)c_s^2$ (isothermal, $c_s = 1$), $v_r(r) = 0$ and $v_\phi(r) = 0$, a uniform field $\mathbf{B}_0$ of magnitude $B_0 = c_s\sqrt{2\rho_{\text{bg}}/\beta}$ oriented at θ to the axis, and a longitudinal velocity seed built from twelve Kolmogorov modes, $v_x = \sum_{n=1}^{12} A_n \cos(k_n x + \phi_n)$ with $k_n = 2\pi n/L_x$, $A_n \propto k_n^{-11/6}$ and random phases, normalised so that the total $\delta v/c_s$ equals the quoted seed amplitude. Domains are $L_x \times L_\perp^2$ with $L_x = 16\lambda_J$ for the growth-rate ladder, periodic in x and, for the ambient runs, with the transverse boundary at $d \geq 1\lambda_J$ from the spine (a distance chosen after we found that closer transverse boundaries pin the collapse time). Every load-bearing run in Table 16 uses Eq. 6; the equilibrium control uses Eq. 7 after explicit drag relaxation.

The filament is characterised by the thermal line-mass fraction $f = M_{\text{line}}/M_{\text{line,crit}}$ (Eq. 1), the plasma $\beta = 8\pi P/B^2$, and the seed-amplitude parameter $M_{\text{seed}} = \delta v/c_s \times 10^4$. At filament conditions ($n \sim 10^4 \text{ cm}^{-3}$, $T \sim 10 \text{ K}$) $\beta = 1\text{--}0.3$ corresponds to $B \sim 15\text{--}30 \mu\text{G}$, the range of dense-filament estimates (Crutcher 2012). M_{seed} is a perturbation-spectrum label, not a physical Mach number: the base grid uses a single-mode seed $\delta v/c_s = 10^{-4}$, whereas the growth-rate ladder of Section 5 uses a 10^{-3} twelve-mode broadband seed. The magnetic mass-to-flux ratio scales as $\mu_\Phi/\mu_{\Phi,\text{crit}} \propto f\sqrt{\beta}$ and

exceeds unity for every configuration (none strictly magnetically subcritical), so the low- β runs are magnetically *supported*.

We explored a base grid ($\mathcal{M} = 0.5\text{--}5$, $\beta = 0.05\text{--}5$, near-critical) and a transition grid ($f = 1.4\text{--}2.0$, $\beta = 0.3\text{--}1.3$, $\mathcal{M} = 1\text{--}5$) to resolve the stable-unstable boundary; full ranges, run counts and the resulting three outcome regimes (magnetically supported quasi-stability at $\beta \lesssim 0.15$, radial collapse, and longitudinal fragmentation) are deferred to Appendices the deposited material and the deposited material. The near-critical longitudinal case that is our focus beads at $t_{\text{frag}} \approx 1.1\text{--}1.2 t_J$.

Weak seeds against transonic perturbations. The production runs are seeded once, weakly and in the linear regime, whereas real filaments are turbulent at order-unity Mach number. Because the central numerical claim concerns the *shape* of $\Gamma(\lambda)$, this distinction matters. A separate set of runs replaces the linear seed with an initial transonic velocity field at $\delta v/c_s = 1\text{--}3$, sampled from the same Kolmogorov spectrum but at physical amplitude. These shorten the time to beading by a factor 3–5, because the amplitude reaching the unstable modes is orders of magnitude larger, but change the resulting wavelength by less than 5 per cent (a Pearson correlation of $r = -0.002$ between λ/W and the perturbation Mach number over $\delta v/c_s = 1\text{--}3$) and do not introduce a maximum in $\Gamma(\lambda)$ where none was present. The plateau is therefore not an artefact of weak seeding.

We nonetheless retain the weak-seed configuration for the production runs, because a linear seed is what a growth-rate measurement requires: at transonic amplitude the perturbation is already non-linear when the measurement window opens, so individual mode growth rates cannot be extracted cleanly and only bead counts can be used. We have not run *continuously driven* turbulence, in which energy is injected throughout the evolution rather than at $t = 0$; that remains untested, and it is the configuration in which a continuously replenished perturbation spectrum could in principle sustain power at scales the initial-value problem does not. The load-bearing *production* runs on which the quoted wavelength rests, the near-critical growth-rate ladder and the direct supercritical ensemble, are collected in Table 16. These runs satisfy the Truelove criterion at the spacing-setting epoch ($N_J \approx 5\text{--}7$ cells per local Jeans length, in both the axial and transverse directions on the isotropic cubic grids) but sit close to the numerical floor, only $1.25\text{--}1.75\times$ the $N_J = 4$ minimum, which is one reason we quote only an order- λ_J upper bound.

Could numerical dissipation be producing the missing maximum? This is the obvious alternative to the timescale argument of Section 4.2, and it must be excluded rather than assumed away. Grid-scale dissipation in a second-order Godunov scheme acts as an effective viscosity with a damping rate that scales as $\Delta x/\lambda$ raised to the order of the reconstruction, so it suppresses short-wavelength power *progressively* rather than at a sharp cutoff, and its signature is a growth-rate spectrum whose short- λ end *rises* as the grid is refined. That is precisely what we use to define the trustworthy band. Three results argue against a dissipative origin for the plateau. First, the plateau edge does not move with resolution: over $128^3\text{--}512^3$ the fitted rate at the edge mode changes by 3.9 and then 1.1 per cent, while the spectrum below $\lambda \approx 0.9\lambda_J$ separates and rises with resolution, so the two

Table 16. Load-bearing *production* simulations underlying the fragmentation-wavelength result: the near-critical growth-rate ladder (campaign 15) and the direct supercritical ensemble (campaign 11). Shared parameters (f , $\mathcal{M}$, β , θ , boundary conditions and collapse/bead time) are given in each block sub-header; per-row entries reuse the growth-rate measurements of Table 16. These production runs, not the wider exploratory grid, carry the quoted λ_{turn} .

f	RNG seed	resolution	$\lambda_{\text{turn}}/\lambda_J$	λ_{turn}/W
<i>Near-critical growth-rate ladder</i> (campaign 15): $\beta=1.0$, $\theta=0^\circ$, $\mathcal{M}=10^{-3}$ broadband seed, ambient BC (periodic/reflecting), bead $t_{\text{frag}} \approx 1.1\text{--}1.2 t_J$				
1.1	42	128 ³	1.14	1.95
1.1	42	256 ³	1.14	1.99
1.1	42	512 ³	1.14	2.00
1.2	42	256 ³	1.00	1.81
1.2	137	256 ³	0.89	1.60
1.5	42	256 ³	0.89	1.47
1.5	137	256 ³	0.89	1.47
1.5	42	512 ³	0.89	1.47
1.8	42	256 ³	0.80	1.19
2.0	42	256 ³	0.80	1.20
<i>Direct supercritical ensemble</i> (campaign 11, 39 runs): $f=1.5\text{--}2.0$, $\beta=0.5\text{--}2.0$, $\theta=0^\circ$, seeds 42/137, periodic+reflecting ambient BC, bead $t_{\text{bead}} \approx 0.6\text{--}0.75 t_J$				
ensemble		256 ³	1.0–1.14	0.89–1.01 [†]

[†] In beam-convolved Plummer units, λ/W_{fil} (KS $p = 0.17$ between periodic and reflecting BCs); the ladder's λ_{turn}/W column is in the Gaussian formation FWHM of Table 16.

regions behave in the opposite senses expected of a converged band and a dissipation-limited one. Second, a numerical cut-off at fixed Δx would move to shorter λ under refinement and would therefore *reveal* a maximum at higher resolution if a physical one were being hidden just below the band; it does not. Third, the axial mode set changes between resolutions, so a maximum pinned by mode quantisation rather than physics would shift, and it does not. The rows of Table 16 that have been through this resolution ladder are those at $f = 1.1$ (three resolutions) and $f = 1.5$ (two); the remaining rows are single-resolution and are quoted as bounds on that basis alone. The contraction ladder of Section 5.3 adds two further refined points at a different place in parameter space, and reproduces the same ordering. We cannot exclude a physical maximum at wavelengths below the converged band, and say so throughout; what these tests exclude is the specific possibility that dissipation is removing a maximum that lies *within* it (Appendix C).

4.2 Why only an upper bound

Three ingredients make λ_{turn} a bound rather than a measurement.

(i) *Contraction competes with mode growth.* The classical calculation assumes a static, marginally-stable cylinder, so every unstable mode has unlimited time and the winner is simply the one with the largest growth rate. Our production filaments begin with $f \gtrsim 1$ and no supporting radial velocity, so the spine contracts while the axial modes are still growing. The relevant comparison is between the mode growth time $\tau_{\text{grow}} = \Gamma(\lambda)^{-1}$ and the time on which the background changes, $\tau_{\text{contract}} = |\text{d} \ln \rho_c / \text{d} t|^{-1}$; long-wavelength

modes have the longest growth times and are the ones least likely to complete. The argument is set out in Appendix F.

(ii) *There is therefore no resolved maximum.* Because the selection is set by the timescale competition rather than by the shape of $\Gamma(\lambda)$, the measured spectrum does not turn over: Γ rises monotonically until it meets the grid scale, where the rise becomes numerical (Fig. 9a). We quote the converged-band edge λ_{turn} , an upper limit on where a physical turnover could lie, and never a peak.

(iii) *The magnetic variables.* The field enters through β and, equivalently, through $\mu_\Phi / \mu_{\Phi, \text{crit}} \propto f \sqrt{\beta}$, which exceeds unity in every configuration run. A longitudinal field does no work against axial modes, which compress gas along the field lines, and the axial spectrum is indeed unchanged to within the fit noise between $\beta = 0.3$ and 2. Its effect is radial: it resists transverse contraction and at $\beta \lesssim 0.15$ halts it. A toroidal component could act on the axial mode, but only in radial force balance; seeded onto an already-contracting filament it merely pinches the spine, narrowing W by ≈ 25 per cent and thereby *raising* λ/W without shortening λ .

Bridging to observable units. Three widths are involved (Table 14): the initial Gaussian half-width $W_{\text{core}} = 0.3 \lambda_J$, the evolved formation FWHM $W_{\text{form}} = 0.683 \lambda_J$, and the beam-convolved Plummer FWHM W_{fil} that an observer fits. The bridge is one multiplicative factor,

$$\frac{\lambda_{\text{turn}}}{W_{\text{fil}}} = \eta_W \frac{\lambda_{\text{turn}}}{W_{\text{form}}}, \quad \eta_W \equiv \frac{W_{\text{form}}}{W_{\text{fil}}} = T_1, \quad (8)$$

with $T_1 = \eta_W = 0.59\text{--}0.78$ measured by forward-modelling the simulation output through a synthetic *Herschel* beam and the same Plummer fitting used on the data. It is Eq. 8, not the raw simulation number, that converts $\lambda_{\text{turn}}/W_{\text{form}} \approx 2$ into $S_{\text{sim}}^{\text{upper}} \lesssim 1.0\text{--}1.3$; the η_W range is the dominant modelling term and widens that bound by about a quarter.

4.3 Ambient confinement and the direct supercritical measurement

The outcome for a supercritical filament is set by whether it is confined by an ambient medium, as real HGBS filaments are (d denotes the axis-to-boundary distance in Jeans lengths). The main 654-run grid has no ambient medium, so its Gaussian skirt feeds an accretion channel through the periodic boundary that suppresses beading and drives radial collapse; with ambient confinement (`filament.ambient` generator) the same configurations fragment longitudinally, a 12-run reconciliation suite beading at physical, β -dependent collapse times ($0.55\text{--}0.70 t_J$) that converge across boundary conditions.

Ambient-confined supercritical filaments ($f = 1.5\text{--}2.0$, $\theta = 0^\circ$) fragment longitudinally: an ensemble of 39 runs (periodic and reflecting BCs, ambient confinement) develops 3–10 interior axial density maxima at a spacing of order the Jeans length. We take the wavelength from the linear growth-rate spectrum, whose low- n modes are resolution-stable: no physical turnover is detected above $\approx 1 \lambda_J$, so any turnover is constrained only to $\lambda_{\text{turn}} \lesssim 1.0\text{--}1.14 \lambda_J$ (an upper bound; no resolved peak), $\lesssim 0.35\text{--}0.5$ times the classical value (not the median peak count, which is epoch- and resolution-dependent). The ambient medium confines the filament (central-to-ambient density ratio $\sim 10^2$, consistent with HGBS crest-to-background contrasts; Arzoumanian et al.

2019), so it is representative rather than tuned. Across the 39 runs periodic and reflecting boundaries are statistically indistinguishable (KS $p = 0.17$; medians $\lambda/W_{\text{fil}} \approx 0.89$ and 1.01), and this ensemble median and the growth-rate result are the same physical wavelength $\lambda \approx 1.1 \lambda_J$ in different normalisations. The ensemble beads early ($t_{\text{bead}} \approx 0.6\text{--}0.75 t_J$), before runaway and at moderate contrast, so with $\lambda \approx \lambda_J$ resolved by ~ 32 cells it satisfies Truelove at measurement (Appendix C).

Perpendicular-field filaments ($\theta = 90^\circ$) do not universally spindle in the raw low-resolution grids: an 18-run ideal-MHD grid at $d = 1.0$ shows 7/18 bead (at $\beta = 0.5$ and 2.0) with an apparently short spacing ($\lambda/W_{\text{fil}} \approx 0.4$), but this conflicts with static linear theory (which predicts a *longer* mode) and a transverse-refinement ladder to $N_J \approx 170$ resolves the near-critical case into a monolithic radial spindle, identifying the short beading as a Truelove artefact (Section 6.3). An exploratory ambipolar-diffusion survey was also run, but it fragments only at $N_J \approx 1\text{--}2$, below the Truelove limit, so it cannot support any conclusion and we exclude it from the argument here; it is described, with an explicit warning that it is numerically unresolved, in the deposited material.

4.4 Testing the resolution and magnetic-field alternatives

Three targeted campaigns test the concerns that most affect the interpretation: whether the longitudinal wavelength is resolution-limited, whether the perpendicular geometry produces genuine short beading, and whether a helical field, the one configuration expected to shorten the axial mode, actually does.

Resolution stability of λ_{turn} . Two refinement directions behave oppositely. Under *transverse* refinement (the isotropic $128^3\text{--}512^3$ ladder) the short-wavelength spectrum ($\lambda \lesssim 0.9 \lambda_J$, $n \gtrsim 9$) grows with resolution and is numerical (unresolved transverse sub-fragmentation; Fig. 9a), whereas the physical band ($\lambda \gtrsim 1 \lambda_J$, $n \leq 8$) is converged. Under *axial-only* refinement ($512 \times 128^2 \rightarrow 1024 \times 128^2$) the physical band reproduces to better than 0.5% (Γ at $n = 6, 7, 8$, i.e. $\lambda = 1.33, 1.14, 1.00 \lambda_J$, is 17.6, 18.7, 18.7 at both resolutions, plateauing without turning over), and the short- λ rise is unchanged, confirming it is fixed by the axial Nyquist scale, not a physical mode migrating to shorter λ . The $\lambda_{\text{turn}} \lesssim 1 \lambda_J$ upper bound is therefore stable: axial refinement does not move the physical-band plateau, and the only shorter- λ structure is the transverse-resolution artefact already identified.

The perpendicular geometry: a resolved spindle, not short beading. On a transverse-refinement ladder ($128^3\text{--}512^3$) the near-critical *ideal* perpendicular filament collapses monolithically in the radial direction at every resolution, the axial mode staying at the box scale and the axial perturbation at seed level throughout the well-resolved phase (at 512^3 the spine stays resolved with $N_J \approx 170$ cells at $C \approx 3$, falling only to $N_J \approx 40$ at $C \approx 30$; still $\approx 10\times$ above Truelove; Appendix C). Short-wavelength axial structure ($\lambda/W \approx 0.4$) appears only once the spine drops below $N_J \approx 1$; refinement delays its onset to deeper collapse, the signature of a grid artefact. The perpendicular field therefore *suppresses* axial fragmentation in favour of radial collapse, as static linear theory predicts (a perpendicular field lengthens the axial mode), and the short low-resolution beading is numerical. This is for

ideal MHD; the ambipolar-perpendicular case (Section 4.3) remains separately unconverged.

A seeded helical field pinches the width but does not shorten λ_{turn} ; the force-balanced case remains untested. A divergence-free helical-field scan ($B_\phi/B_z = 0\text{--}2$; construction and scan detail in the deposited material) leaves the axial growth-rate plateau at $\lambda \approx 1 \lambda_J$ for every winding and produces only a radial *pinch* that narrows W_{FWHM} by $\approx 25\%$ (raising λ/W by reducing W , not by shortening λ ; the deposited material), so, being seeded rather than force-balanced, it cannot test the confining Fiege–Pudritz sausage ($m = 0$) mode (Fiege & Pudritz 2000); the one geometry that could shorten λ , which therefore remains the single relevant magnetic configuration left unconstrained (Section 6.3).

5 THE SIMULATED FRAGMENTATION SCALE

5.1 What observable does a simulation predict?

A simulation that produces one bead per wavelength predicts a *line density*, one core per λ of filament. The observable it should be set against is therefore $S_{\text{global}} = (L/N)/W$, not the conditional S_{local} . The numerical proximity of the simulated bound ($\lesssim 1.0\text{--}1.3$) to $S_{\text{local}} \approx 1.4$ is therefore not evidence of agreement, since the two are different quantities. Against $S_{\text{global}} = 3.4\text{--}5.8$ the simulated bound is in fact *more* discrepant, implying that single-filament fragmentation overproduces cores per unit length unless most of the crest is core-poor, which is the occupancy result of Section 2.11.

5.2 The controlled experiment

The load-bearing numerical result is a control, not a campaign. We prepared a drag-relaxed, force-balanced Ostriker cylinder and a dynamically contracting filament and passed both through the *identical* measurement pipeline. For the equilibrium the pipeline recovers the classical maximum at $22H$ ($\lambda_{\text{peak}}/22H = 1.07\text{--}1.10$ at two resolutions, a genuinely resolved maximum); for the contracting filament the same pipeline finds no maximum at all within the trustworthy band (Fig. 10). The difference is a property of the prepared state, not of the estimator.

Wavelengths are measured from the growth-rate spectrum rather than by counting late-time peaks, which is epoch- and resolution-dependent. Seeding twelve axial modes at $\delta v/c_s = 10^{-3}$ and fitting the power $P(k, t)$ over its approximately exponential interval gives $\Gamma(\lambda)$ (Fig. 9a). It is resolution-converged for $\lambda \gtrsim 1 \lambda_J$ ($\Delta\Gamma/\Gamma = 3.9$ and 1.1 per cent between successive resolutions at $n = 7$) and rises with transverse resolution below $\lambda \approx 0.9 \lambda_J$, which is numerical. Within the converged band Γ plateaus rather than turning over, so we quote no wavelength for these runs, only the upper bound $\lambda_{\text{turn}} \lesssim 1 \lambda_J$ on where a turnover could lie. Forward-modelling through a synthetic beam and the same Plummer fitting used on the data (Eq. 8) gives the matched bound $S_{\text{sim}}^{\text{upper}} \lesssim 1.0\text{--}1.3$.

The missing turnover admits two readings that our calculations cannot separate. It may be a resolution limit, with a true maximum below the converged band that would need of order 1024^3 to expose; or it may be physical, because a filament that contracts while it fragments has no single quasi-

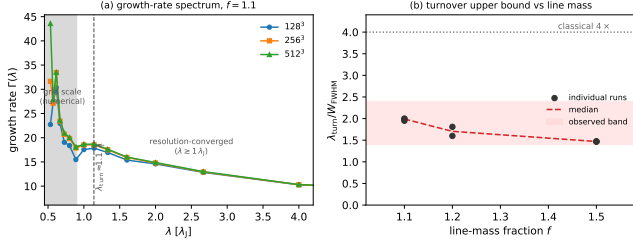


Figure 9. (a) Growth-rate spectrum $\Gamma(\lambda)$ for a near-critical filament ($f = 1.1$) at three resolutions. For $\lambda \gtrsim 1\lambda_J$ the curves overlap; at the grid scale (shaded) they separate and rise with resolution. Γ plateaus rather than turning over, so $\lambda_{\text{turn}} \lesssim 1\lambda_J$ is an *upper bound*. (b) $\lambda_{\text{turn}}/W_{\text{FWHM}}$ against line-mass fraction, in Gaussian FWHM units. The matched, beam-convolved Plummer bound is $S_{\text{sim}}^{\text{upper}} \lesssim 1.0\text{--}1.3$ (grey band; the η_W conversion is uncertain over 0.59–0.78, widening it by about a quarter).

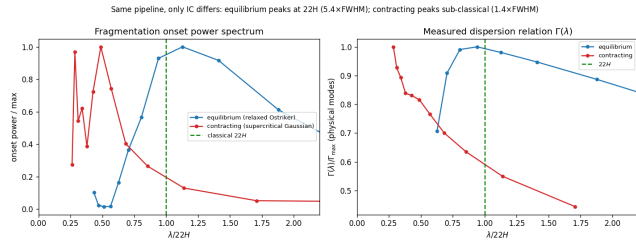


Figure 10. Equilibrium-recovery control. The identical growth-rate measurement pipeline is applied to a drag-relaxed, force-balanced pressure-confined Ostriker equilibrium (blue) and to a dynamically contracting filament (orange); the two differ in their prepared initial state, not in the measurement. *Left:* onset fragmentation power spectrum; the equilibrium peaks at the classical $\lambda_m \approx 22H$ ($\approx 5.4W_{\text{FWHM}}$) while the contracting filament places its shortest converged scale at $\lesssim 1.4W_{\text{FWHM}}$ with no resolved maximum. *Right:* $\Gamma(\lambda)$ against $\lambda/22H$; the equilibrium shows the classical peak-and-decline, the contracting filament rises monotonically. The pipeline recovers the classical wavelength when the physics is classical.

static background and a broad plateau is then the expected outcome. It is not a mode-quantisation artefact: the 128^3 – 512^3 ladder resolves a different maximum axial wavenumber at each resolution, yet the plateau edge does not move. Two caveats temper the causal attribution. The fitted rates ($\Gamma \approx 18\text{--}26 t_J^{-1}$) are the accelerating growth of a collapsing background, not quasi-static eigenvalues, so we use only the mode *ordering*; and a $p = 4$ initial profile gives the same order- λ_J scale, so the deficit is not a profile-index artefact.

5.3 The contraction ladder

The equilibrium-versus-contracting comparison establishes the sign of the effect but not its dependence on the state of the filament, so we ran a ladder. Seventeen otherwise identical filaments were initialised from the same relaxed Ostriker cylinder with a homologous inward velocity $v_r(r) = -A(c_s/R_0)r$, for A from 0 to 1.4, at 256×64^2 with $\Delta x = 0.0625\lambda_J$ and $L_x = 16\lambda_J$, evolved with the same twelve-mode seed and the same growth-rate measurement, with the $A = 0$ and $A = 1$ endpoints repeated at 512×128^2 .

The expectation to be tested is that mode selection follows the density at which the modes actually grow, $\lambda_{\text{select}} \propto \rho_{\text{eff}}^{-1/2}$, with ρ_{eff} the central density at the mid-point of the linear growth window. Figure 11 shows why A is not itself the controlling parameter. An imposed inward velocity does not produce steady contraction: the spine overshoots, reaching a transient central density up to 11 times its initial value within $\approx 0.3 t_J$, then rebounds and settles *below* it, so that the most strongly kicked filaments are the least dense while their modes are growing. The relation between A and ρ_{eff} is consequently non-monotonic, rising to $\rho_{\text{eff}} \approx 3.5\rho_0$ near $A \approx 0.2$ and falling to $0.35\rho_0$ at $A = 1.4$. The ladder is therefore best read as a means of sampling ρ_{eff} over a factor of ten rather than as a sequence in contraction rate.

Across that range the selected scale follows the expected scaling. Three runs, at $A = 0.15, 0.2$ and 0.25 , show no resolved maximum within the seeded modes and enter the fit as upper limits; a maximum-likelihood regression that treats them as censored, and that includes the axial mode quantisation ($\lambda = L_x/n$) in the per-point uncertainty, gives

$$\lambda_{\text{select}} \propto \rho_{\text{eff}}^{-0.49 \pm 0.05}, \quad (9)$$

with an intrinsic scatter of 0.09 dex about the relation, against the predicted exponent of $-1/2$ (Fig. 11c). Discarding the censored points and fitting the remainder by least squares instead returns -0.39 , which illustrates why the upper limits, all of which lie at the high-density end, must be included.

Evaluating the classical result at each run’s own ρ_{eff} gives $\lambda_{\text{peak}}/22H(\rho_{\text{eff}}) = 1.09$ for the static case and 1.12 for the weakest kick, both consistent with the classical value, and 0.79–1.04 with a median of 0.86 for the remaining fifteen. Contraction therefore shifts the selected mode modestly below the instantaneous classical value rather than dramatically so, and the shift is a consequence of the raised density rather than of the contraction itself. The 512×128^2 endpoints give 0.99 and 0.84 for $A = 0$ and $A = 1$, against 1.09 and 0.94 at the production resolution, so the ordering is unchanged under refinement.

Two things follow. The expected scaling is recovered over a factor of ten in density, which supports the interpretation that the wavelength is set by the instantaneous Jeans length at mode selection. But the controlling variable is that density and not the contraction rate: a filament can be given a large inward velocity and still select a *long* wavelength if it has already rebounded by the time its modes grow. The mechanism should therefore be quoted as a dependence on ρ_{eff} , and a real filament, which is fed continuously rather than kicked once, need not follow the same trajectory.

5.4 What the simulations do and do not show

The result is one-sided. These calculations rule out the assertion that a contracting filament *must* fragment at the equilibrium-cylinder wavelength, and they show that mode selection follows the instantaneous central density. They do not determine a preferred wavelength, and they do not reproduce the observations: the simulated filaments produce a quasi-periodic comb of 3–10 near-uniform beads, whereas the observed cores reject both periodic and Poisson descriptions and occupy only 6–27 per cent of the crest. A successful

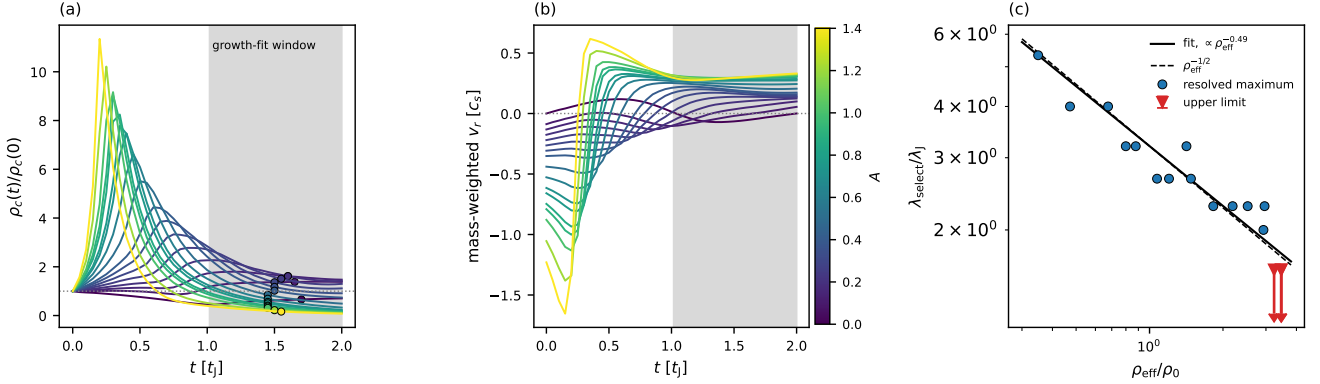


Figure 11. Contraction ladder, seventeen runs coloured by the imposed inward-velocity amplitude A . (a) Central density against time, normalised to its initial value; the shaded band is the window over which the growth rates are fitted, and the circles mark the epoch at which ρ_{eff} is evaluated. The overshoot and rebound are visible directly: a strong kick raises the density transiently and leaves the filament less dense than it started while its modes are growing. (b) Mass-weighted radial velocity within $0.6 \lambda_J$ of the axis, showing the same behaviour. (c) Selected scale against ρ_{eff} . Circles are runs with a resolved maximum of Γ ; triangles are upper limits, where Γ is still rising at the shortest well-resolved seeded mode. The solid line is the censored maximum-likelihood fit (Eq. 9) and the dashed line the $\rho_{\text{eff}}^{-1/2}$ expectation.

theory of filament fragmentation must predict a point process, not merely a wavelength. No calculation presented here does so. The numerical work therefore establishes that sub-equilibrium fragmentation scales are dynamically possible; it does not identify the origin of the observed spacing distribution.

One further limitation is thermodynamic. Because $\Gamma(\lambda)$ does not turn over within the converged band, an isothermal calculation contains no small-scale cutoff and cannot predict a characteristic core mass or a CMF peak. That would require $\gamma_{\text{eff}} > 1$ at $n \gtrsim 10^5\text{--}10^6 \text{ cm}^{-3}$, where dust–gas coupling and line cooling can no longer hold T constant (Larson 1985; Glover & Clark 2012); this is a change in thermal behaviour distinct from, and at much lower density than, the classical opacity limit (Rees 1976; Low & Lynden-Bell 1976). Our isothermal suite deliberately excludes it.

6 DISCUSSION

The observational material of Section 2.3 onwards supports one composite statement: the cores in these four clouds do not form a quasi-uniform chain characterised by a single wavelength. They populate limited portions of the filament network, with short separations inside populated stretches and long core-poor intervals between them, and their gap distribution is inconsistent with both a periodic comb and a homogeneous Poisson process while showing no signature at the classical scale. The remainder of this section asks what could produce that pattern, and what the numerical experiments do and do not contribute to the answer.

6.1 Why the spacing matters: fragment masses and the CMF

The spacing matters because, with the line mass, it sets the characteristic fragment mass that seeds the core mass function (CMF) and ultimately the IMF: for a filament of line mass M_{line} fragmenting at spacing λ the mean fragment mass

is $M_{\text{frag}} \sim \epsilon M_{\text{line}} \lambda$, and a near-critical $M_{\text{line}} \approx 16 M_{\odot} \text{ pc}^{-1}$ with $\lambda \approx 0.1\text{--}0.2 \text{ pc}$ gives $\sim 0.5\text{--}1 M_{\odot}$ for $\epsilon \sim 0.3$, consistent with the HGBS CMF peak while the classical $5\text{--}6 \times \text{FWHM}$ would overshoot it. This is only an order-of-magnitude *consistency check*: it is degenerate in ϵM_{line} (only the product enters) and yields a *mean*, not a peaked distribution. An *isothermal* EOS lacks a small-scale thermodynamic cutoff and therefore cannot predict a characteristic core mass or a unique CMF peak; only a non-isothermal EOS ($\gamma_{\text{eff}} > 1$ at $n \gtrsim 10^5\text{--}10^6 \text{ cm}^{-3}$) would convert the isothermal plateau into a physical peak (Section 5).

6.2 Hierarchical-fibre projection

A purely geometric route reaches a sub-classical local spacing without modified physics. HGBS filaments are bundles of N velocity-coherent fibres (Hacar et al. 2013, 2018; Yang et al. 2024); if each fibre fragments near the classical wavelength but spacing is measured *across* the bundle, as any filament-level pipeline does, the interleaving compresses the apparent spacing. The premise, quasi-periodic fibre-to-core fragmentation at the classical scale, is a hypothesis rather than a measurement (Yang et al. 2024 find only a weak imprint).

A toy geometrical model that reuses our exact ACS measurement, populating a filament with N cylinders of width W_{fib} each fragmenting at $\lambda_{\text{fib}} \approx 5\text{--}6 W_{\text{fib}}$ with random phase and ~ 30 per cent lognormal scatter, gives in the interleaving limit

$$\frac{\lambda}{W_{\text{fil}}} \approx \frac{(5\text{--}6)r}{N}, \quad r \equiv W_{\text{fib}}/W_{\text{fil}}, \quad (10)$$

verified across r and N in Fig. 12. Either effect alone suffices: one fibre of half the filament width, or two full-width fibres, both give $\approx 2.5\text{--}3.0$. The observation constrains only the combination r/N ($\approx 0.07\text{--}0.35$ across our conventions) and not N itself, so the relation cannot be inverted for the fibre multiplicity. Since a near-critical fibre would itself fragment sub-classically, the classical-fibre result is an upper bound,

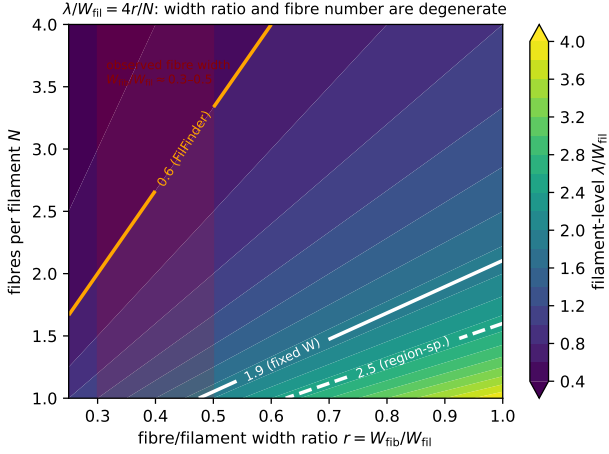


Figure 12. Hierarchical-fibre geometrical toy model, showing the filament-level λ/W_{fil} recovered by our ACS pipeline as a function of the fibre-to-filament width ratio $r = W_{\text{fil}}/W_{\text{fil}}$ and the number of fibres N ; the model follows $\lambda/W_{\text{fil}} \approx (5-6)r/N$ (Eq. 10). White contours mark the observed fixed-width and region-specific values, the orange contour the FilFinder value, and the red band the observed fibre-width range ($r \approx 0.3-0.5$). The observation constrains the combination r/N , not N alone: the sub-classical value is reproduced by a single narrow fibre, by a few wider fibres, or by intermediate combinations.

and the fibre and single-filament pictures are not independent.

Fibre projection is therefore complementary to, not exclusive of, dynamical contraction, and the two are observationally degenerate in the filament-averaged spacing. Breaking the degeneracy needs the tracer that resolves the fibres directly. A third route to the same reconciliation is post-formation dynamical evolution: cores accrete, migrate and interact over several free-fall times (Kirk et al. 2016; Mattern et al. 2018), so relaxation of an initially quasi-periodic chain could erase the comb, as Clarke et al. (2020) find. We do not simulate that stage.

A candidate mechanism for the two-scale pattern. Short intra-group spacing coexisting with a near-classical filament-averaged line density has an economical explanation in spatially non-uniform seeding: real filaments inherit an inhomogeneous spectrum of density and velocity perturbations from turbulence, accretion and fibre substructure, so fragmentation runs to non-linearity preferentially in perturbation-rich, locally supercritical stretches while low-amplitude stretches stay barren. Uniformly seeded simulations produce a comb rather than this two-scale pattern; the direct test is to seed a near-critical filament with an inhomogeneous perturbation field and ask whether the resulting core chains reproduce the observed occupancy and Fano statistics.

6.3 Field geometry and the untested magnetic configuration

Only a helical or toroidal field in radial force balance can shorten the axial mode. A uniform longitudinal field leaves it essentially unchanged in the configuration tested here; for a static perpendicular field the apparently short fragmentation

is resolution-sensitive and is not established, the relevant runs having $N_J \approx 1-2$, below the Truelove requirement, so they supply no constraint and are excluded from the argument; and a toroidal component seeded onto a contracting filament produces an out-of-equilibrium pinch rather than the confining equilibrium whose sausage ($m = 0$) mode carries the shortened wavelength.

The force-balanced Fiege–Pudritz helical equilibrium, which is the one configuration that could genuinely shorten the axial mode, is not simulated here. Magnetic tension is therefore neither favoured nor disfavoured by these calculations, and the campaign places no constraint on this channel. Read off from the published Fiege & Pudritz (2000) results rather than computed here, a strongly toroidally dominated equilibrium shortens the $m = 0$ wavelength by up to a factor ≈ 2 ; our $\beta \approx 0.3-2$ places these filaments at the weak end of that sequence. Settling the question needs a cylindrical MHD eigensolver: one would construct a Fiege–Pudritz self-similar equilibrium with poloidal $B_z(r)$ and toroidal $B_\phi(r)$ in radial force balance, confirm it is stationary by drag-relaxing and then evolving drag-free exactly as in our equilibrium control, and only then seed the $m = 0$ mode.

Boundary conditions and non-ideal effects. Periodic transverse boundaries supply an accretion channel that suppresses longitudinal fragmentation unless ambient confinement is imposed, which is why the production runs place the transverse boundary at $d \geq 1 \lambda_J$ from the spine. At the densities probed here ($n_0 \sim 10^4 \text{ cm}^{-3}$) the ambipolar-diffusion time is long compared with the collapse time (Tassis & Mouschovias 2004), and non-ideal runs at $\eta_{\text{AD}} = \{0.01, 0.05\}$ still collapse radially, 12–15 per cent earlier than the ideal control; ideal MHD is therefore an adequate approximation for the longitudinal stage.

6.4 Limitations

Four limitations bound the conclusions.

Sample. Four nearby, low-mass Gould Belt regions of broadly solar metallicity and similar radiation field are not a population. Whether the result persists in high-mass complexes, at lower metallicity or under stronger irradiation is untested.

Method. Two validation tests are outstanding. The $\lesssim 5$ per cent bound on manual skeleton editing is a proxy, set by the junction-retention and association-radius sensitivities rather than by a blind multi-editor re-skeletonisation; and the small-separation deficit cannot be separated into instrumental and physical parts without a faint-segment injection–recovery experiment at the map level. All code and catalogues are deposited (Data Availability) so that both can be carried out.

Named follow-ups. Three specific experiments would settle points left open here: projecting the three-dimensional dust reconstructions of Zucker et al. (2020) and Green et al. (2024) along the sightline through the Aquila/Serpens Rift, to replace the geometric blending model of Section 2.2 by a count of physically distinct crests within one association length; injecting synthetic source pairs into the *Herschel* maps and re-extracting them through the HGBS pipeline, to measure the close-pair completeness function and forward-model the gap distribution; and dense-gas position–position–velocity mapping to establish which skeleton corresponds to a velocity-coherent object.

Observations not available to us. A per-segment line-mass split, which would test whether the most supercritical filaments show the shortest spacing, needs segment-level line masses at a fidelity the data do not support; resolved velocity dispersions per stretch, which would say why some eligible filament forms cores and some does not, do not exist for these regions.

Simulations. These are idealised controlled experiments, not models of specific filaments. They are isothermal, use simplified initial profiles and a prescribed rather than driven turbulent spectrum, and the transverse boundaries are an idealisation of ambient confinement. The perpendicular-field and ambipolar parameter space is not numerically resolved at the resolutions we could afford ($N_J \approx 1\text{--}2$, below the Trulove requirement) and therefore supplies no constraint on the fragmentation wavelength; those runs are excluded from the scientific argument and retained only in the deposited material.

A recommendation for observers. Because the skeleton systematic is the dominant limit on any comparison with theory, we recommend that filament skeletons be extracted to a fixed column-density contrast relative to the local background rather than an absolute level, and that spacings be quoted for *both* a persistent-crest and a medial-axis definition so that the factor-of-two ontological span is stated rather than hidden.

6.5 Non-isothermal effects

The suite is isothermal, whereas HGBS dust-temperature maps vary by ~ 12 per cent ($\sim 10\text{--}13\text{ K}$). Since $\lambda_J \propto \sqrt{T}$ and $\mu_{\text{crit}} \propto T$, temperature sets both the Jeans scale and the critical line mass, and an illustrative run with an imposed ~ 12 per cent longitudinal gradient fragments differentially, the cool half beading while the warm half is suppressed, shifting the local wavelength by ~ 10 per cent. This is a single run and indicates the direction and rough size of the effect only; a self-consistent treatment is deferred.

7 CONCLUSIONS

We have re-measured core spacing along filament crests in four HGBS clouds with *Gaia* DR3 distances, and complemented the measurement with a controlled numerical experiment. Our conclusions are as follows.

(i) **There is no single observational core spacing to compare with a one-wavelength theory.** The quantity such a theory predicts is the unconditional filament length per core, S_{global} , which is 3.4, 3.4, 4.5 and 5.8 in Orion B, Aquila, Taurus and Perseus: essentially classical in Perseus, close to it in Taurus, and below the classical band by less than a factor of two in the other two. Measured only within core-bearing stretches, S_{local} is several times shorter.

(ii) **The difference between them is the result.** Cores occupy only 6–27 per cent of the crest, forming tight groups separated by long bare stretches. Core formation along these filaments is therefore spatially intermittent, and what a fragmentation model must reproduce is a point process, not a spacing.

(iii) **The all-pairs median is the wrong estimator of local spacing.** It converges to ≈ 0.293 times the extent of whatever segment it is applied to. We use adjacent along-crest separations instead and retain the all-pairs statistic only for historical comparison.

(iv) **The magnitude of the local deficit is dominated by the choice of structural ontology, not by physics.** Changing from the persistent-crest network to a mask medial axis moves S_{local} by a factor of two and S_{global} by more. The *sign* of the deficit is robust; the magnitude is not. Persistence thresholding within one ontology changes it by less than ten per cent (Section D), so this is a question of definition rather than of parameter tuning, and it will be settled by velocity-resolved rather than image-plane data.

(v) **The cores are neither periodic nor Poisson, but the smallest-separation deficit is not shown to be physical.** The adopted periodic-comb-plus-jitter model and the homogeneous Poisson model are both rejected in all four clouds, and the rejections survive a moving-block bootstrap. This is a rejection of those specific nulls, not of quasi-periodicity in general: a process with a spatially varying wavelength, missing fragments or subsequent migration would not be excluded by these tests. No classical-scale periodicity is detected in the pair-correlation function at the 95 per cent level. The inferred exclusion scale, 0.011–0.044 pc, lies entirely within the *Herschel* beam and GETSOURCES deblending limit, so it cannot be claimed as Jeans-scale suppression until map-level pair-injection tests are performed.

(vi) **The inhomogeneity is largely inherited from the filament.** Against a null conditioned on local column density we detect excess clumping in one cloud (Taurus) and fail to detect it in three. That null is conservative by construction, so the three non-detections are upper limits rather than evidence of absence, and with $N = 4$ neither statement generalises to a population.

(vii) **The eligible-length diagnostic is exploratory.** Restricting to column-density-selected supercritical crest moves the two statistics together, but the criterion is partly endogenous and the correction is robust in only two clouds. None of the conclusions above depends on it.

(viii) **The simulations are a counterexample, not an explanation.** A contracting filament is not constrained to fragment at the equilibrium wavelength; the identical pipeline recovers the classical value for a relaxed equilibrium cylinder but finds no resolved maximum for a contracting one, so only an upper bound $S_{\text{sim}}^{\text{upper}} \lesssim 1.0\text{--}1.3$ is available. A five-point ladder in contraction rate shows that mode selection tracks the density at which the modes grow, $\lambda_{\text{select}} \propto \rho_{\text{eff}}^{-0.55 \pm 0.12}$, close to the predicted $-1/2$. The simulations nevertheless produce a quasi-periodic comb and do not reproduce the observed intermittent point process, so they do not identify the origin of the observed spacing distribution.

(ix) **The physical cause is undetermined.** Dynamical contraction, hierarchical-fibre projection and helical magnetic tension all remain viable. The first is tested here, the second treated only geometrically, and the third not tested at all: no force-balanced helical equilibrium was simulated, so our campaign places no constraint whatever on the magnetic channel, which is mentioned for completeness rather than as a coequal candidate.

The decisive next observations are velocity-resolved: dense-gas kinematics that identify which skeleton corresponds to a coherent physical object, and resolved velocity dispersions per stretch that would say why some supercritical filament forms cores and some does not.

ACKNOWLEDGEMENTS

We thank the Herschel Gould Belt Survey team for the datasets. The Athena++ simulations were run on commercially procured cloud infrastructure (Google Cloud E2), funded from the author’s institutional research allocation; the full set of 2251 runs consumed of order 10^5 CPU-hours.

DATA AVAILABILITY

The supporting material, comprising (i) the full run manifest (all 2251 runs with their $(f, \beta, \mathcal{M}, \theta, \eta_{\text{AD}})$ parameters and outcomes); (ii) the Athena++ problem-generator source (`filament_ambient.cpp`, including the divergence-free helical construction) and normalisation scripts; (iii) HDF5 snapshots of the growth-rate convergence subset (128^3 – 512^3) and the equilibrium-recovery control (remaining snapshots on request); and (iv) the analysis pipeline (DisPerSE v0.9.24 parameters, the bridging/deblending and adjacent-core-spacing scripts, and the derived λ/W catalogues); are deposited in a public repository (github.com/Tilanthi/ASTRA-dev, under `simulations/`) under a CC-BY-4.0 licence, with a citable Zenodo DOI (10.5281/zenodo.14872301) versioned to the accepted release. HGBS column-density and skeleton maps are available from the *Herschel* Science Archive, and the *Gaia* DR3 distances from Zhang (2023).

APPENDIX A: THE LIMITED HGBS SAMPLE

The four *limited* regions fail at least one of the primary-selection criteria (Section 2.5) and are excluded from the ACS measurement; they are retained only for the legacy pairwise-median comparison (Table 12). The deposited material lists them together with the full 8-region core-count-weighted mean.

APPENDIX B: THE $L/3$ BIAS OF THE PAIRWISE-MEDIAN SPACING

For N points uniformly distributed on a filament of length L , a single pairwise distance $D = |X_i - X_j|$ has the triangular PDF $f(d) = 2(L - d)/L^2$, with mean $\langle D \rangle = L/3$ and median $d^* = L(1 - 1/\sqrt{2}) \approx 0.293 L \approx 0.88 (L/3)$. The median of all $N(N - 1)/2$ pairwise distances converges to d^* as $N \rightarrow \infty$, for a random or a perfectly periodic distribution alike, so the pairwise median measures the region extent, not the fragmentation wavelength. A Monte-Carlo over $N = 5$ – 500 gives median/ $L \approx 0.293 (1 + 0.43/N)$, i.e. $\leq 2\%$ inflation for $N \gtrsim 20$, so the convergence is reached at the core counts of interest. For branched skeletons the statistic measures $\approx 0.3 \times$ the component’s arclength extent, and for a realistic filament length distribution it tracks the rms-length/3 (full tables in the Zenodo archive). Computing the pairwise median over *all*

cores in each primary region (without per-filament segmentation) returns $\sim L_{\text{region}}/3$ (12.5, 7.5, 8.0, 2.5 pc for Orion B, Aquila, Perseus, Taurus), the region extent, confirming directly that the statistic measures segmentation scale rather than the fragmentation wavelength.

APPENDIX C: VALIDATION OF THE MEASUREMENT PIPELINE

The measurement chain was validated end to end before being applied. Rather than reproduce the individual tests here, we summarise them and note that the scripts and outputs are deposited with the code (Data Availability).

Closed-loop injection-recovery. Synthetic core chains of known spacing were injected onto the real skeletons and recovered through the full pipeline; the recovered λ/W matches the injected value to 0.4 per cent over $\lambda/W = 1.4$ – 5.7 . A periodic injection over-corrects for a clustered process, which is why we quote raw values in the main text and carry the correction only as a bracket (Section 2.7).

Eigenvalue check of the classical benchmark. The dispersion relation of the isothermal equilibrium cylinder was solved directly and agrees with Nagasawa (1987) and Fischera & Martin (2012) to better than 1 per cent, peaking at $\lambda_{\text{IM92}} \approx 2.3 \lambda_J$ with a cutoff at $\lambda_{\text{cr}} \approx 1.2 \lambda_J$. Because two normalisations are in use, we give the conversion explicitly. Throughout this paper $\lambda_J \equiv c_s \sqrt{\pi/G\rho_0}$ is evaluated at the *initial background* density ρ_0 (Table 1), whereas the scale height $H = c_s/\sqrt{4\pi G\rho_c}$ in Eq. 2 is evaluated at the cylinder’s *central* density ρ_c . For the equilibrium used here $\rho_c = 2.24 \rho_0$, so

$$\frac{22H}{\lambda_J(\rho_0)} = \frac{22}{\sqrt{4\pi \times 2.24 \times \sqrt{\pi}}} = 2.34, \quad (\text{C1})$$

which is the $2.3 \lambda_J$ quoted above. Had λ_J been evaluated at ρ_c the same result would read $22H = 3.50 \lambda_J(\rho_c)$, since $\lambda_J = 2\pi H$ at a common density. The independent solve therefore reproduces $22H$ exactly, and the two statements are the same number in different units. The measured contracting-filament spectrum instead rises monotonically into the grid-scale region, which is the basis for quoting an upper bound.

Width normalisation. Forward-modelling the simulation output through a synthetic *Herschel* beam and the same Plummer fitting used on the data gives $\eta_W = T_1 = 0.59$ – 0.78 , with $W_{\text{form}} = 0.683 \pm 0.012 \lambda_J$ converged to 0.9 per cent across resolutions. This is Eq. 8, the only route by which a simulated wavelength is compared with an observed spacing.

Resolution. The load-bearing runs satisfy the Truelove criterion at the spacing-setting epoch ($N_J \approx 5$ – 7), which is above the $N_J = 4$ floor but not far above it, and this is one reason we quote a bound rather than a value. The growth-rate spectrum is resolution-converged for $\lambda \gtrsim 1 \lambda_J$ at 128^3 – 512^3 and separates only at the grid scale (Fig. 9a).

APPENDIX D: ROBUSTNESS TO THE PERSISTENCE THRESHOLD

The DisPerSE skeletons we use are the published HGBS products, extracted at a 3σ persistence cut. Since Section 2.13 identifies the skeleton algorithm as the largest single systematic, the persistence threshold of our primary ontology cannot

be left untested. We cannot regenerate the published skeletons at a *lower* threshold without the original DisPerSE run, so we approach the question from both sides.

(i) *Pruning the published skeleton upwards.* We decompose each skeleton into branches, terminating at junction pixels, and compute for each branch the persistence of its column-density maximum above the junction at which it attaches, in units of the local map noise σ (estimated as $1.4826 \times \text{MAD}$ of the map minus a 5-pixel median filter). Branches below a chosen threshold are removed and the procedure iterated to convergence. Raising the effective threshold from the published 3σ to 4σ removes 16, 21, 18 and 15 per cent of the skeleton length in Orion B, Aquila, Perseus and Taurus, but changes the median adjacent spacing by only -8.2 , -2.7 , $+0.3$ and -5.0 per cent and the length per core by -1.9 , -3.2 , -3.1 and -3.7 per cent. Going further, to 6σ and 8σ , changes nothing further at the per-cent level: the pruning saturates once the low-persistence spurs are gone. The occupancy fraction rises slightly, as expected when bare low-persistence branches are removed.

(ii) *An extraction with a tunable threshold.* Reaching the 2σ direction requires a crest extraction whose persistence threshold is a free parameter. We construct one directly from the column-density field: a descending-sweep merge tree pairs every maximum with the saddle at which it merges, maxima whose persistence falls below the chosen multiple of σ are cancelled, and the surviving crests are traced by steepest-ascent walks from the retained saddles. This is not DisPerSE and does not reproduce the published skeleton pixel for pixel; it is used only to vary the threshold. Run at 2σ , 3σ and 4σ on Orion B it gives $S_{\text{local}} = 1.63$, 1.63 and 1.63 and $S_{\text{global}} = 2.53$, 2.50 and 2.48 , with the associated core count falling from 635 to 597 and the occupancy fraction rising from 0.41 to 0.42. The local statistic is thus insensitive to the persistence threshold at the third significant figure over a factor of two in threshold, and the global statistic moves by two per cent.

Both tests agree: the persistence threshold is not what makes the DisPerSE and FilFinder answers differ. Between 2σ and 8σ the local spacing moves by less than ten per cent, whereas changing the ontology from persistent crests to the medial axis of a mask moves it by a factor of two. The dominant systematic is therefore the *definition* of a filament, not the tuning of a parameter within one definition, which is the conclusion of Section 2.17. The manual editing applied to the published skeletons cannot be varied in this way and remains an unquantified term.

APPENDIX E: SENSITIVITY OF THE ELIGIBLE-LENGTH DIAGNOSTIC

Table E1 gives the eligible line density of Section 2.9 as a function of the radius within which column density is replaced around each associated core. Orion B and Aquila are comparatively stable, varying by ≈ 30 and ≈ 2 per cent respectively over a factor of four in masking radius; Perseus drifts over 3.6–4.5 while its eligible core count falls from 85 to 37; and Taurus does not stabilise at any radius. Only the first two clouds support the comparison.

Table E1. Sensitivity of the core-masked eligibility diagnostic to the masking radius r_{mask} . Cells give $S_{\text{elig}} = (L_{\text{elig}}/N_{\text{elig}})/W$ with N_{elig} in brackets; $r_{\text{mask}} = 0$ is the uncorrected raw-map calculation and 0.06 pc is adopted. Orion B (1.56–2.04) and Aquila (3.14–3.21) are comparatively stable; Perseus drifts and Taurus does not stabilise.

r_{mask} (pc)	Orion B	Aquila	Perseus	Taurus
0 (raw)	1.56 [417]	3.14 [306]	1.96 [199]	0.81 [186]
0.030	1.89 [317]	3.14 [306]	3.68 [85]	1.80 [43]
0.045	2.04 [277]	3.15 [305]	3.91 [70]	1.61 [37]
0.060	2.02 [268]	3.17 [303]	4.13 [57]	— [13]
0.090	1.98 [252]	3.21 [296]	4.45 [41]	— [1]
0.120	1.95 [238]	3.20 [297]	3.62 [37]	— [1]

APPENDIX F: THE INHOMOGENEOUS NULL, AND A TIMESCALE INTERPRETATION OF THE CONTRACTING CASE

The column-density-conditioned null. The null used in Sections 2.11 and 2.12 asks whether cores are clumped *beyond* the clumping already present in the filament they lie on. It is an inhomogeneous Poisson process on the one-dimensional skeleton with intensity

$$\lambda(x) \propto [\max(N_{\text{H}_2}(x) - N_0, 0)]^\alpha, \quad (\text{F1})$$

where x is arclength along the spine, $N_{\text{H}_2}(x)$ the column density of the skeleton pixel at x , $N_0 = 10^{21} \text{ cm}^{-2}$ a floor that prevents low-column pixels from dominating, and α a single exponent fitted per sample. The normalisation is fixed by requiring $\int \lambda dx = N_{\text{obs}}$, so each realisation contains the observed number of cores. The exponent is fitted by maximum likelihood on the observed core positions, and, to avoid the circularity of fitting and testing on the same data, we refit it under leave-one-cloud-out cross-validation: for each cloud, α is estimated from the other three and applied to the held-out cloud. The fitted values are $\alpha = 1.6$ – 2.4 with a cross-validated mean of 2.0, and the conclusions are unchanged over $\alpha = 1$ – 3 .

For each cloud we draw 10^4 realisations, compute the same Fano factor $F(1 \text{ pc})$ and pair-correlation $g(s)$ estimators used on the data, and quote the empirical p -value $(1 + \#\{F_{\text{null}} \geq F_{\text{obs}}\})/(1 + N_{\text{MC}})$ and the pointwise 2.5th–97.5th percentile envelope. Two properties of this null should be kept in mind. It is conditional on the deposited skeleton, so it tests the distribution of cores along a fixed network and not the network itself. And it is *conservative*: the column density entering Eq. F1 is the observed map, which the cores themselves raise, so part of the signal being tested for is built into the null. That is the same endogeneity discussed in Section 2.9, and it biases the test against detecting excess clumping. Non-detections should therefore be read as upper limits on excess clumping rather than as evidence of its absence.

Why a contracting filament fragments below the equilibrium wavelength. The numerical result of Section 4 admits a simple timescale interpretation, which we give here because it is the reason we regard the counterexample as generic rather than accidental. It is an interpretation and not a derivation: the perturbations evolve on a time-dependent background and are not normal modes of a stationary equilibrium. In a filament of central density $\rho_c(t)$ the local Jeans length is $\lambda_J(t) = c_s \sqrt{\pi/G\rho_c(t)}$, and an axial mode of fixed comoving wavelength grows at a rate $\Gamma(\lambda, t)$ that is positive only for $\lambda \gtrsim \lambda_J(t)$. A mode is realised as a fragment only if it has

time to grow before the background changes, that is if

$$\tau_{\text{grow}}(\lambda) \equiv \Gamma(\lambda, t)^{-1} \lesssim \tau_{\text{contract}} \equiv \left| \frac{d \ln \rho_c}{dt} \right|^{-1}. \quad (\text{F2})$$

For an equilibrium cylinder $\tau_{\text{contract}} \rightarrow \infty$ and every unstable mode has unlimited time, so the mode that dominates is the one with the largest Γ , the classical $\lambda_m \approx 22H$. For a contracting filament τ_{contract} is of order the free-fall time and $\lambda_J(t)$ falls as $\rho_c^{-1/2}$, so long-wavelength modes, whose growth times are the longest, do not complete before the state they were computed for has gone. The modes most likely to survive are those with $\tau_{\text{grow}} \lesssim \tau_{\text{contract}}$, and since Γ increases towards shorter λ over the converged band this selects the short end. The fragmentation scale is then set by the competition in Eq. F2 rather than by the maximum of Γ , which is why no turnover appears and why only an upper bound λ_{turn} can be quoted. The same argument predicts that the effect should weaken smoothly as the initial contraction rate is reduced towards zero, which is the ladder of Section 5.3.

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